

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

Gradient-Routed Alignment with Cross-tokenizer Experts
for Multimodal Understanding

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

HÀ NỘI - 2026

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



LE XUAN AN

**Gradient-Routed Alignment with Cross-tokenizer Experts
for Multimodal Understanding**

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Giáo sư Võ Đình Hiếu

HÀ NỘI - 2026

VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Le Xuan An

**Gradient-Routed Alignment with Cross-tokenizer Experts
for Multimodal Understanding**

BACHELOR'S THESIS

Major: Computer Science

Advisor: Dr. Nguyen Van Son

Co-Supervisor: Professor Vo Dinh Hieu

HANOI - 2026

Lời cảm ơn

Assurance

I hereby affirm that this thesis is my own original work. I have not submitted it previously for any academic or professional award.

I completed this thesis through my own efforts, with guidance and support from Dr. Nguyen Van Son and Professor Vo Dinh Hieu, in partial fulfillment of the graduation requirements. I confirm that all information, data and findings presented here are original and free from plagiarism.

I have acknowledged every source that I consulted and included each one in the references. I accept full responsibility for the accuracy and integrity of this thesis under the regulations of VNU University of Engineering and Technology.

Hanoi, Ngày 7 tháng 5 năm 2026

Student

Le Xuan An

Tóm tắt

Khóa luận này nghiên cứu chứng cất tri thức đa giáo viên không đồng nhất từ một tập teacher VLM đóng băng sang một student VLM. Bài toán khó vì các teacher đến từ nhiều họ mô hình khác nhau, dùng tokenizer khác nhau, và không teacher nào hữu ích như nhau trên mọi mẫu. Vì vậy, cách lấy trung bình cố định giữa các teacher thường kém hiệu quả khi độ hữu ích của từng teacher thay đổi theo từng mẫu.

Khóa luận đề xuất Gradient-Routed Alignment with Cross-tokenizer Experts (GRACE). Tên gọi này nhấn mạnh hai thành phần chính của phương pháp: routing để chọn tập teacher phù hợp cho từng mẫu và alignment khác tokenizer thông qua hàm mất mát trie-Wasserstein trên cây byte dùng chung. Khung này xử lý trực tiếp ba vấn đề thực tế: lệch tokenizer, khác biệt thể mạnh giữa các teacher và bất ổn sau bước routing. Phương pháp kết hợp hàm mất mát trie-Wasserstein trên cây byte dùng chung, teacher gate phụ thuộc vào student, sparse top- k routing và cơ chế tinh chỉnh trọng số GRACE dựa trên gradient. Gate trước hết chọn một tập teacher nhỏ cho từng mẫu. Sau đó, GRACE so sánh gradient KD của từng teacher với gradient CE có giám sát của student, làm mượt điểm số bằng exponential moving average rồi áp dụng trọng số KD cuối cùng trong tập teacher đã được route.

Thực nghiệm trên TextVQA, DocVQA và ChartQA cho thấy GRACE luôn tốt hơn finetuning trực tiếp và các baseline chứng cất đã so sánh. Trong thực nghiệm, student được khởi tạo từ SmolVLM-500M và SmolVLM-256M. Trong thiết lập bốn teacher được báo cáo, GRACE đạt 69.38, 73.11 và 80.20 với SmolVLM-500M, và 60.73, 60.48 và 72.32 với SmolVLM-256M. Với SmolVLM-500M, mức cải thiện tương ứng so với mô hình gốc là 22.26%, 22.83% và 27.71%. Phương pháp cũng thu hẹp 55.03% khoảng cách tới teacher trên TextVQA, 39.53% trên DocVQA và 55.34% trên ChartQA.

Từ khóa: mô hình thị giác-ngôn ngữ, chứng cất tri thức, đa giáo viên, mixture-of-experts routing, chứng cất Wasserstein khác tokenizer

Abstract

We study heterogeneous multi-teacher distillation for visual language models (VLMs), where supervision is transferred from a pool of frozen teachers to a student model. The setting is difficult because teachers come from different model families, use different tokenizers and provide complementary supervision across samples, which makes fixed teacher averaging unreliable.

We present GRACE, short for Gradient-Routed Alignment with Cross-tokenizer Experts, as a framework designed to address three failure modes in this setting: tokenizer mismatch, heterogeneous teacher specialization and unstable post-routing weighting. In this thesis, the frozen heterogeneous teachers are treated as tokenizer-specific experts and the final teacher assignment is gradient-routed because routed teacher weights are refined using agreement between each teacher’s KD gradient and the supervised CE gradient. To the best of our knowledge, it is the first distillation framework to employ a shared tree-metric OT for heterogeneous multi-teacher VLM distillation. It combines a trie-Wasserstein KD loss on a shared byte trie, a teacher routing network, sparse top- k routing and gradient-agreement refinement. After the routing network selects a sparse teacher subset for each sample, GRACE compares each teacher’s KD gradient with the student’s supervised CE gradient, smooths the resulting agreement scores with an exponential moving average and converts them into final KD weights over the routed teacher set.

In the reported TextVQA, DocVQA and ChartQA experiments, GRACE improves over direct finetuning and the compared distillation baselines. The experimental student models are SmolVLM-500M and SmolVLM-256M. In the four-teacher setting, GRACE reaches 69.38, 73.11 and 80.20 on SmolVLM-500M and 60.73, 60.48 and 72.32 on SmolVLM-256M. For SmolVLM-500M, this corresponds to gains of 22.26%, 22.83% and 27.71% over the original student. The method also closes 55.03% of the original teacher gap on TextVQA, 39.53% on DocVQA and 55.34% on ChartQA. These results support GRACE as an effective framework for heterogeneous teacher-to-student VLM distillation in the evaluated VQA setting.

Keywords: visual language model, knowledge distillation, multi-teacher distillation, mixture-of-experts routing, cross-tokenizer Wasserstein distillation

Table of Contents

Lời cảm ơn	
Assurance	i
Tóm tắt	ii
Abstract	iii
Table of Contents	iv
List of Figures	vii
List of Tables	ix
List of Abbreviations	x
Chapter 1 Introduction	1
1.1 Background and Motivation	1
1.2 Related Work	6
1.2.1 Response-Based Knowledge Distillation	7
1.2.2 Cross-Tokenizer and Transport-Based Distillation	7
1.2.3 Layer Distillation	7
1.2.4 Multi-Teacher Distillation	8
1.2.5 Research Gap	8
1.3 Problem Statement and Research Objectives	8
1.4 Research Questions	9
1.5 Thesis Contributions	9
1.6 Scope and Assumptions	10
1.7 Thesis Organization	11
Chapter 2 Preliminaries	13
2.1 Transformer-Based Vision-Language Models	13

2.1.1	Transformer Basics	13
2.1.2	Vision Encoder and Multimodal Fusion	13
2.2	Knowledge Distillation	14
2.2.1	Teacher-Student Learning	14
2.2.2	Temperature-Scaled Output Distributions	15
2.2.3	Answer-Token Distillation	15
2.3	Heterogeneous Teacher Distribution Matching	16
2.3.1	ULD as a Prior Baseline	16
2.3.2	Wasserstein Distance and Discrete Optimal Transport	17
2.3.3	Tree-Structured Mass Matching	19
2.4	Multi-Teacher Distillation and Routing	19
2.4.1	Per-Teacher KD Loss	19
2.4.2	Learned Routing and Load Balancing	20
2.4.3	Gradient Conflict and Teacher Agreement	22
Chapter 3	Proposed Method	24
3.1	Method Overview and Problem Formulation	24
3.1.1	Notation	26
3.1.2	Training Flow	27
3.2	Trie-Wasserstein Cross-Tokenizer Distillation	28
3.2.1	Supervised Answer Tokens	28
3.2.2	Shared Byte Trie	28
3.2.3	Trie-Wasserstein Objective	29
3.2.4	Sparse Support and Residual Tail Edge	30
3.3	Teacher Routing Network	32
3.3.1	Router Architecture	32
3.3.2	Top-k Routing	33
3.3.3	Router Entropy and Z-Loss	34
3.4	Gradient-Agreement Refinement	35
3.4.1	Gradient Agreement Scoring	35
3.4.2	Agreement Scores and Active Set	35
3.4.3	Warmup, EMA Smoothing and Score Weights	36
3.4.4	Weight Blending and Fallback	36
3.4.5	Per-Sample KD Aggregation	38
3.5	Final Objective	38
Chapter 4	Evaluation Methodology	39

4.1	Experimental Setup	39
4.2	Datasets	39
4.2.1	Dataset Descriptions	39
4.2.2	Dataset Size and Splits	40
4.3	Training Configuration	40
4.4	Evaluation Metrics	43
4.5	Baselines and Compared Methods	44
4.6	Implementation Details	47
4.6.1	Offline Teacher-Logit Preparation	48
4.6.2	Remote Logit Streaming During Training	49
Chapter 5	Experiment Results	51
5.1	Main Comparison Results	51
5.1.1	Baseline Results	51
5.1.2	Overall Comparison	51
5.2	Representation Similarity Analysis	63
5.3	Ablation Study	65
5.4	Efficiency Analysis	67
5.5	Discussion	68
5.6	Limitations and Threats to Validity	69
	Conclusion and Future Work	71
Chapter A	Closed-Form Trie-Wasserstein Derivation	72
A.1	Setup	72
A.2	Lower Bound	73
A.3	Achievability	73
	References	75

List of Figures

1.1	CLIP-style vision-language pretraining with a shared image-text embedding space [1, 2].	1
1.2	Scaling-law trends from Kaplan et al.	2
1.3	SmolVLM memory comparison from the Hugging Face release.	3
1.4	Representative VLM model scales.	3
1.5	TinyLLaVA vs. LLaVA-1.5-7B benchmark comparison.	4
1.6	Best proprietary vs. open-source frontier results on three benchmarks.	4
1.7	Low-bit quantization can sharply reduce VLM accuracy.	5
1.8	Teacher-student knowledge distillation.	6
2.1	Transformer and ViT building blocks.	14
2.2	Representative architecture families of modern vision-language models.	15
2.3	ULD distillation procedure.	17
2.4	Tokenizer vocabulary overlap example.	17
2.5	Geometric behavior of Wasserstein distance.	18
2.6	Tree-structured and matrix views of discrete optimal transport.	20
2.7	Sparse routing with expert capacity constraints.	21
2.8	Sparse routing and expert specialization in MoE.	22
2.9	Gradient conflict in a multi-objective setting.	23
3.1	GRACE architecture.	25
3.2	Shared-trie view of tokenizer-mismatched matching for document/receipt and doc/rece	29
3.3	Sparse trie-Wasserstein support and residual tail mass.	32
3.4	Teacher routing network.	33
3.5	Top- k routing dynamics in GRACE.	34
3.6	Representative KD weighting regimes.	37
4.1	Representative dataset examples.	40
4.2	R2 bucket used for teacher-logit storage.	48
4.3	Training with remote teacher logits.	49

5.1	Grouped comparison of original, best-baseline and GRACE results.	56
5.2	DEX-AR overview.	57
5.3	Representative TextVQA attention example.	58
5.4	Training-curve comparison for uniform weighting and GRACE.	59
5.5	Representative collapse and non-collapse routing examples.	60
5.6	Final-layer CKA heatmaps across two benchmarks.	64
5.7	Vision-layer CKA heatmaps across two benchmarks.	64

List of Tables

3.1	Main symbols used in Chapter 3.	26
3.2	Illustrative ULD and trie-Wasserstein comparison.	32
4.1	Dataset split sizes for the three benchmarks.	40
4.2	Main training protocol for the active distillation launch path.	41
4.3	Baseline groups used in the evaluation chapter.	44
4.4	Engineering settings for the active logit-streaming pipeline.	49
5.1	Original SmolVLM student results.	52
5.2	Teacher and reference-model results.	52
5.3	TextVQA main comparison.	53
5.4	DocVQA main comparison.	54
5.5	ChartQA main comparison.	55
5.6	Relative gains of GRACE on SmolVLM-500M.	61
5.7	Gap-to-teacher analysis for SmolVLM-500M.	61
5.8	DocVQA component ablation.	66

List of Abbreviations

Abbr.	Full Term	Description
ANLS	Average Normalized Levenshtein Similarity	Evaluation metric used for DocVQA.
API	Application Programming Interface	Interface used to query model outputs in black-box settings.
bf16	Brain Floating Point 16	Low-precision floating-point format used for efficient mixed-precision training.
BLIP-2	Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models	A modular vision-language model with a Q-Former bridge.
CCA	Canonical Correlation Analysis	Classical similarity analysis method related to the interpretation of CKA scores.
CE	Cross-Entropy	Supervised loss used for classification or distillation objectives.
CKA	Centered Kernel Alignment	Representation similarity measure used to compare hidden layers between teacher and student models.
ChartQA	Chart Question Answering	Benchmark for question answering over charts.
CLIP	Contrastive Language-Image Pre-training	Dual-encoder vision-language pretraining framework.
CNN	Convolutional Neural Network	Standard visual backbone architecture used as a comparison point to ViT.
DocVQA	Document Visual Question Answering	Benchmark for question answering on document images.
EOS	End of Sequence	Special token that marks the end of a generated answer span.

Abbr.	Full Term	Description
FFN	Feedforward Network	Transformer sublayer applied after attention.
fp16	Floating Point 16	Half-precision floating-point format used in mixed-precision training or inference.
GRACE	Gradient-Routed Alignment with Cross-tokenizer Experts	Proposed multi-teacher distillation method in this thesis.
GPU	Graphics Processing Unit	Parallel hardware accelerator used for training and serving deep learning models.
SHM-CKA	Hidden-State Matching with CKA	Baseline distillation method that aligns teacher and student hidden representations using CKA.
InfoNCE	Information Noise-Contrastive Estimation	Contrastive objective for aligning image and text representations.
JSON	JavaScript Object Notation	Lightweight structured data format used for the LLaVA-style training files in this thesis.
JS	Jensen-Shannon	Symmetric divergence measure between probability distributions.
KD	Knowledge Distillation	Transfer of knowledge from a teacher model to a student model.
KL	Kullback-Leibler	Measure used to compare probability distributions.
KLD	Kullback-Leibler Divergence	Divergence-based objective widely used in distillation.
LLaVA	Large Language and Vision Assistant	Instruction-tuned projector-based vision-language model.
LLM	Large Language Model	Language model with large parameter scale and strong reasoning ability.
LN	Layer Normalization	Normalization method used to stabilize Transformer training.
MLP	Multi-Layer Perceptron	Feedforward projection block used in Transformers and classifiers.

Abbr.	Full Term	Description
MMD	Maximum Mean Discrepancy	Similarity metric used for feature-level alignment.
MoE	Mixture of Experts	Sparse architecture that routes tokens to selected experts.
MSA	Multi-Head Self-Attention	Attention block that models token interactions in Transformers.
MTKD	Multi-Teacher Knowledge Distillation	Distillation setting that uses more than one teacher model.
NLP	Natural Language Processing	Field concerned with computational understanding of language.
NLS	Normalized Levenshtein Similarity	String similarity score used in DocVQA evaluation.
OCR	Optical Character Recognition	Extraction of text from images or scanned documents.
OT	Optimal Transport	Distribution-matching perspective used to motivate tokenizer-agnostic distillation.
TWD	Tree-Wasserstein Distance	Tree-structured optimal-transport distance used as background for the trie-Wasserstein loss in this thesis.
PE	Positional Encoding	Representation of token order in Transformer models.
PLN	Parametric Layer Normalization	Layer normalization variant with trainable scale and bias.
Q-Former	Querying Transformer	Bridging module used by BLIP-2 to connect vision and language features.
QA	Question Answering	Task of generating answers to natural language questions.
ReLU	Rectified Linear Unit	Common non-linear activation function.
RMSNorm	Root Mean Square Layer Normalization	Lightweight normalization method based on root mean square statistics.
RoPE	Rotary Positional Embedding	Positional encoding method based on rotational transformations.

Abbr.	Full Term	Description
TF32	TensorFloat-32	NVIDIA tensor-core numeric format used to accelerate matrix operations during training.
TextVQA	Text Visual Question Answering	Benchmark for question answering on natural images containing text.
ULD	Universal Logit Distillation	Tokenizer-agnostic distillation loss based on ranked probability distributions.
ViT	Vision Transformer	Transformer-based model for image representation learning.
VLM	Vision-Language Model	Model that processes visual and textual inputs together.
VLM-Eval-Kit	Vision-Language Model Evaluation Kit	Toolkit used in this thesis to run standardized multimodal evaluation.
VNU	Vietnam National University	Parent university system of VNU University of Engineering and Technology.
VQA	Visual Question Answering	Task of answering questions grounded in visual inputs.
ZeRO	Zero Redundancy Optimizer	Memory-optimization technique used by DeepSpeed to reduce optimizer-state replication during large-model training.

Chapter 1

Introduction

1.1 Background and Motivation

Vision-Language Models (VLMs) process images and text together for tasks such as image captioning, visual question answering, document understanding and instruction following [2–6]. Within this setting, the core task is to transfer strong multimodal capabilities from stronger VLM teachers to a student VLM that remains practical under deployment constraints. In this thesis, *multimodal understanding* refers to reading text in images, understanding the input question and performing basic reasoning over visual and textual evidence [7–9].

Many modern VLM families still build on the basic vision-text pretraining pattern shown in Figure 1.1. This pattern first learns a shared space for image and text signals, then later VLM variants add instruction tuning, cross-attention modules or language-model decoders on top of that foundation.

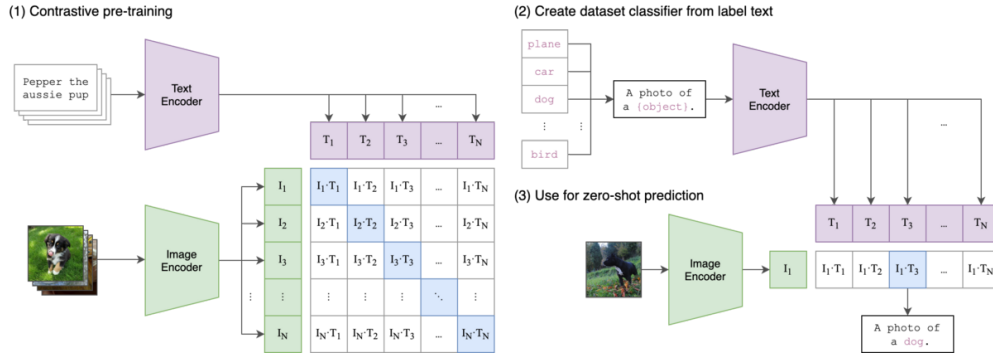


Figure 1.1: CLIP-style vision-language pretraining with a shared image-text embedding space [1, 2].

Although alternative architectures exist for vision-language modeling, including state-space models (SSMs) and Mamba-based designs [10], this thesis focuses on **Transformer methods**. All teachers and students considered in this work use Transformer backbones for both vision and language processing.

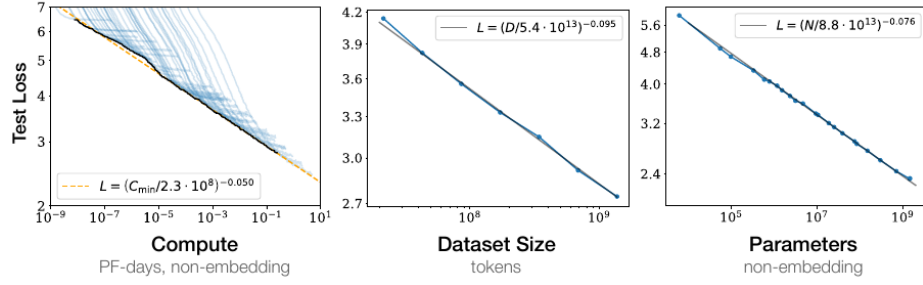


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

Figure 1.2: Scaling-law trends from Kaplan et al. [11].

Scaling-law results help explain why the strongest teacher models tend to be large, since performance generally improves as model size, training data and compute increase. Kaplan et al. quantify this scaling pattern, while Hoffmann et al. show that compute-optimal training still requires increasing model size and data tokens [11, 12]. Larger models therefore remain attractive for accuracy but costly to train and deploy because they require more GPU memory, longer inference time and larger serving infrastructure. This creates a practical gap between the models that achieve the strongest benchmark scores and the models that can be used under limited deployment budgets.

High-performing VLMs often incur high memory use, latency and deployment cost [3, 4, 13, 14]. This constraint is more pronounced in VLMs because visual tokens increase the effective sequence length and Transformer attention scales quadratically with sequence length [5, 15, 16]. Therefore, deployment pressure motivates teacher-to-student distillation, but the method itself is not tied to a fixed parameter budget. This setting is relevant when the student must fit on limited hardware, serve many requests cheaply or act as the deployable endpoint after training with stronger teachers.

Deployment pressure is illustrated by the SmolVLM memory comparison in Figure 1.3 and by the model-scale comparison in Figure 1.4.

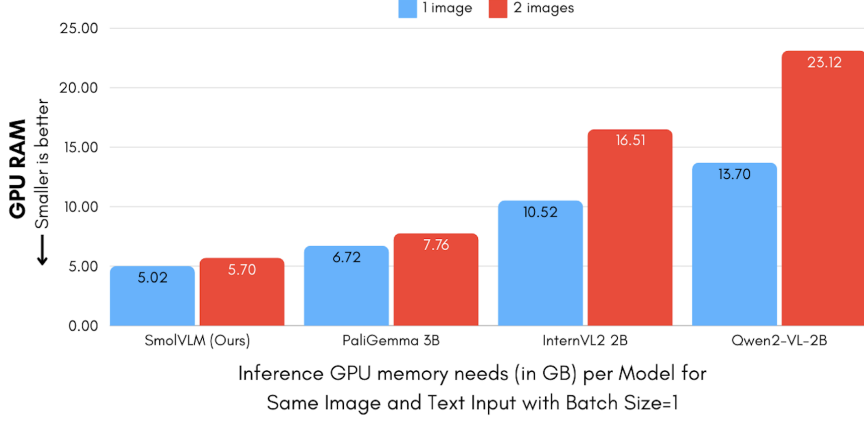


Figure 1.3: Inference GPU memory usage from the SmolVLM release blog [16, 17].

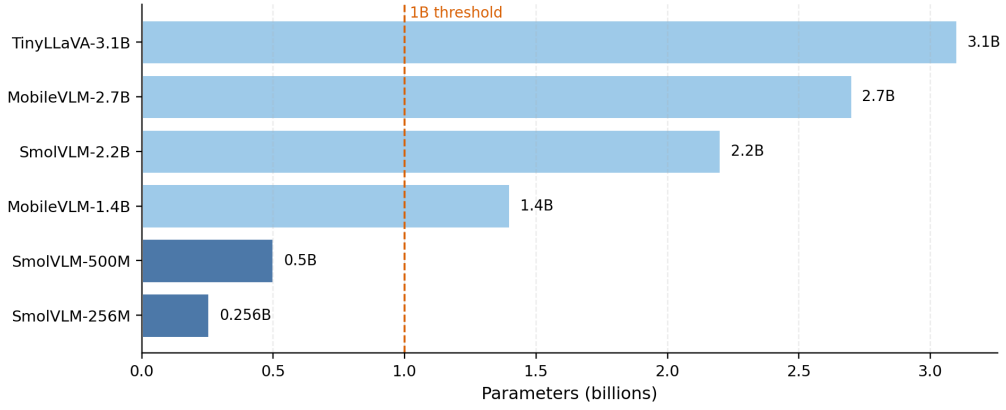


Figure 1.4: Representative VLM model scales used to contextualize the student models in the experiments [13, 14, 16].

This deployment setting is especially important for *multimodal tasks*, where the model must read image text, understand the question and perform basic reasoning. Therefore, this thesis evaluates on TextVQA, DocVQA and ChartQA [7–9].

Although lightweight multimodal models can already be competitive (Figure 1.5), closing the remaining gap to higher-capacity systems still requires more effective knowledge transfer from stronger teachers.

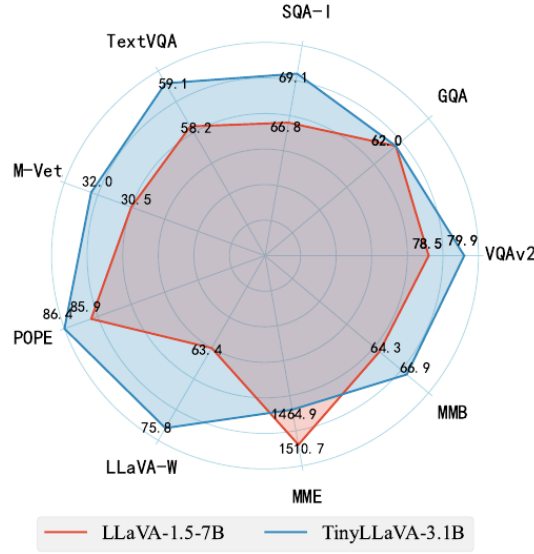


Figure 1.5: TinyLLaVA vs. LLaVA-1.5-7B benchmark comparison from [14].

On these three benchmarks, the best open-source models are already close to the best proprietary models and outperform them in two cases (Figure 1.6).

Open-source frontier models are already close to proprietary leaders

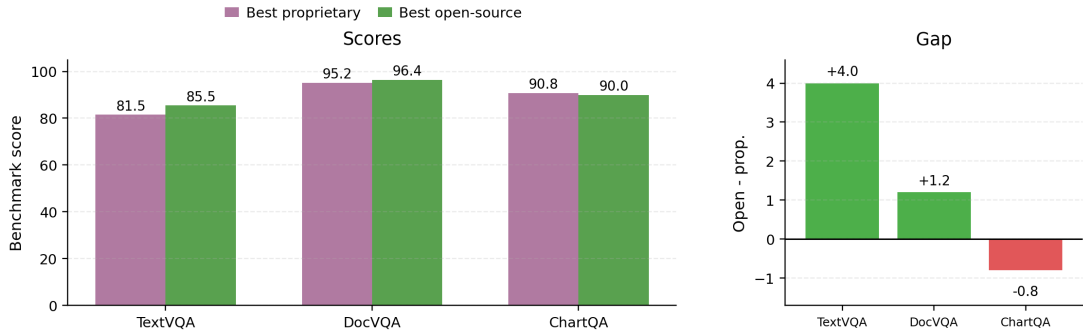


Figure 1.6: Representative proprietary vs. open-source frontier results on TextVQA, DocVQA and ChartQA [18–20].

Figures 1.2 through 1.6 summarize the central tension: scaling improves performance and competitive teachers are available, yet practical deployment often requires efficient student models. Knowledge distillation (KD) bridges this gap by transferring supervision from high-performing teachers to student models [21, 22].

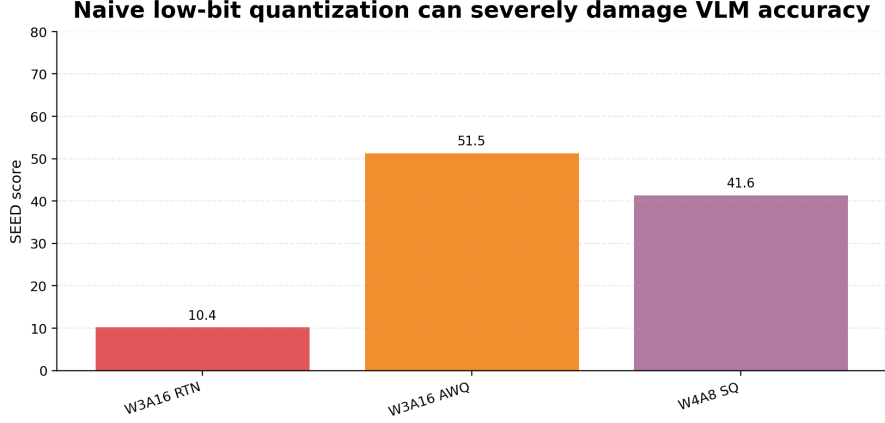


Figure 1.7: SEED accuracy of LLaVA-OneVision-7B under several low-bit quantization settings, reconstructed from MBQ Table 4 [31].

Because competitive open-source VLMs are accessible as teachers, *white-box KD* becomes especially attractive. In *black-box KD*, the student can only query an external teacher service and read its returned predictions. In *white-box KD*, the student can also use teacher logits and intermediate states [23, 24]. This additional access enables the trie-Wasserstein KD loss used here, as well as baseline methods such as ULD and hidden-state matching, which are not usable in a pure black-box setting [22, 25–27].

Another major compression approach is *quantization*, which compresses an already trained model by reducing numerical precision. It can cut memory and improve inference speed substantially, so it is a serious practical alternative to KD [28–30].

In VLMs, however, naive low-bit quantization can hurt multimodal accuracy [30, 31]. For this reason, we treat quantization as a complementary deployment step, not a substitute for better transfer during training.

Figure 1.7 illustrates this limitation: lowering precision can reduce VLM accuracy sharply, so quantization is treated as a deployment complement, not the main transfer mechanism.

Figure 1.8 shows the basic teacher-student KD setting, which this work extends to heterogeneous multi-teacher distillation.

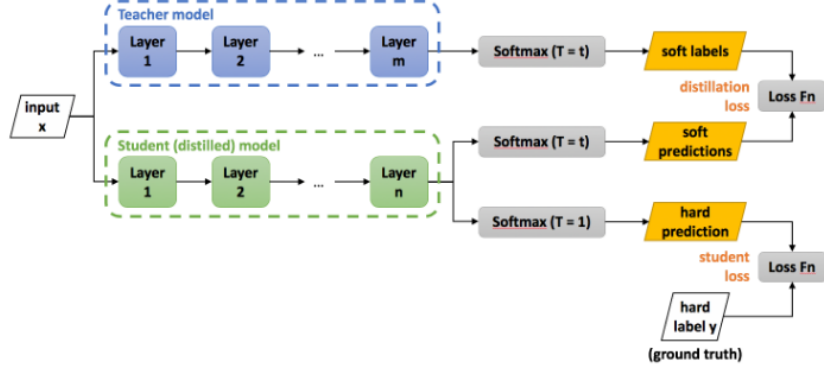


Figure 1.8: Generic teacher-student knowledge distillation [21, 32].

Heterogeneous multi-teacher distillation is harder than the standard single-teacher setting because the teachers may differ in architecture, tokenizer, vocabulary and task-specific strength. The candidate teachers can come from distinct VLM families with mismatched tokenizers, output vocabularies and strengths that vary across samples [25, 33, 34]. A naive uniform average can therefore combine useful and harmful teacher signals for the same sample [22, 33, 34]. This motivates selective multi-teacher distillation, where teacher assignment depends on the input sample.

We present **Gradient-Routed Alignment with Cross-tokenizer Experts (GRACE)**, a heterogeneous multi-teacher distillation framework that treats frozen VLM teachers as tokenizer-specific experts, routes a sparse teacher subset for each sample and aligns teacher-student output distributions through trie-Wasserstein KD with gradient-agreement refinement.

1.2 Related Work

The most relevant literature can be grouped by transfer mechanism: matched-vocabulary logits distillation, tokenizer-mismatched distillation, representation matching and multi-teacher selection.

1.2.1 Response-Based Knowledge Distillation

Classical KD matches softened teacher and student output distributions, usually under a shared tokenizer or matched output space [21, 22]. When teacher and student share the same tokenizer, the common choice is a Kullback-Leibler (KL) divergence between softened logits [21]. In multimodal systems, prior work has also transferred reasoning behavior or representation knowledge into vision-language models [6, 35, 36]. These studies show that lightweight VLMs can benefit from teacher supervision, but they usually assume one teacher or a matched output space. They therefore do not directly address heterogeneous teachers with mismatched tokenizers.

Quantization is a separate compression line that reduces numerical precision after or during training [28–31]. It helps deployment, but it does not solve the transfer problem studied here.

1.2.2 Cross-Tokenizer and Transport-Based Distillation

Tokenizer-mismatched KD becomes necessary when teacher and student vocabularies do not match. ULD is a direct baseline in this setting because it compares sorted probability mass instead of token IDs [25]. More recent work also studies tokenizer-agnostic distillation through transport-based formulations [26, 27, 37].

The weakness of ranked-mass matching is that it discards relations between decoded strings. Two pairs with the same probability shape can therefore receive the same loss even when one pair shares a long prefix and the other does not. Tree-structured transport keeps that information by comparing mass on shared branches of a hierarchy [26, 38]. This motivates objectives that match teacher and student distributions by shared substructure instead of token ID.

1.2.3 Layer Distillation

Another line of work transfers intermediate representations instead of only final logits. Similarity measures such as CKA provide a tokenizer-independent way to compare hidden states across layers [39]. This makes representation matching attractive when teacher and student architectures differ. However, hidden-state alignment alone does not determine which teacher should receive the largest weight for a given sample and does not resolve conflicts among several teachers.

1.2.4 Multi-Teacher Distillation

Multi-teacher KD has been studied with fixed averages, stochastic teacher selection and meta-learned weighting rules [33, 34, 40]. These methods show the value of multiple teachers, but fixed or meta-learned weights can still mix useful and harmful supervision when teacher utility varies across samples.

Sparse routing and load balancing are well established in mixture-of-experts models [41–43]. Gradient-based filtering also appears in distillation through alignment-style objectives [44]. Together, these ideas suggest that a multi-teacher student should first narrow the teacher pool and then check whether the selected teachers provide gradients that agree with the supervised objective.

1.2.5 Research Gap

Prior work leaves three gaps that must be handled together. **First**, heterogeneous teacher and student VLMs often use different tokenizers and output vocabularies, so direct token-index KL is not valid. **Second**, teacher usefulness changes across image-question pairs, which means that a fixed teacher average can dilute useful supervision with weaker or conflicting signals. **Third**, even after a router selects a candidate teacher set, the routed KD gradients may still disagree with the supervised objective. These gaps motivate a method that combines tokenizer-agnostic distribution matching, sample-dependent teacher selection and post-routing gradient refinement.

1.3 Problem Statement and Research Objectives

Given a pool of frozen teacher VLMs and a student VLM, the goal is to learn a heterogeneous multi-teacher multimodal distillation strategy that preserves task performance while reducing deployment cost. The strategy must satisfy four requirements: it must remain valid when teacher and student architectures, tokenizers and output vocabularies differ, assign teachers adaptively for each sample, keep routing sparse through per-sample top- k teacher selection and suppress routed teachers whose KD direction conflicts with the supervised objective.

Gradient-Routed Alignment with Cross-tokenizer Experts (GRACE) addresses these requirements as a selective multi-teacher distillation method for VLMs. In this name, “Experts” refers to the frozen heterogeneous teacher VLMs, while “Routed Alignment” refers to selecting a sparse teacher subset and aligning teacher-student output distributions through trie-Wasserstein KD.

1.4 Research Questions

This thesis is organized around three research questions that evaluate GRACE from the perspectives of predictive performance, component contribution and computational efficiency.

- **RQ1: Performance Comparison.** How does GRACE compare with direct fine-tuning, same-family logits KD, ULD and heterogeneous multi-teacher baselines on TextVQA, DocVQA and ChartQA?
- **RQ2: Component Analysis.** How much do trie-Wasserstein KD, sparse top- k routing and gradient-agreement refinement each contribute to the final performance?
- **RQ3: Efficiency Analysis.** What execution-time and computational costs are introduced by GRACE?

These questions structure the experimental chapter: the main benchmark comparison answers RQ1, the ablation study answers RQ2 and the runtime and cost analysis answers RQ3.

1.5 Thesis Contributions

The thesis makes four main contributions that together define, implement and evaluate **Gradient-Routed Alignment with Cross-tokenizer Experts (GRACE)** for heterogeneous teacher-to-student VLM distillation.

Heterogeneous multi-teacher VLM distillation. The thesis formulates the problem of distilling a student VLM from teachers with mismatched architectures, tokenizers and output vocabularies. This setting captures the practical case where strong teachers come from different model families and cannot be compared with the student through standard same-vocabulary logits KD.

Gradient-routed teacher alignment. The method addresses cases where KD gradients conflict with the supervised task objective by refining routed teacher weights according to their agreement with the CE gradient. This makes teacher weighting both sample-dependent and objective-aware after sparse routing has selected the candidate teacher set.

Tree-metric optimal transport for KD. The thesis introduces trie-Wasserstein distillation, which maps decoded token strings into a shared UTF-8 byte trie and computes KD through tree-metric subtree-mass imbalance. This preserves token-string structure while avoiding dense vocabulary-level optimal transport.

Evidence against representation-only matching. The experiments evaluate hidden-state matching with CKA as a heterogeneous teacher baseline and show that representation similarity alone is not sufficient for effective distillation in this setting. This motivates the combination of output-level matching, routing and gradient-aware refinement.

1.6 Scope and Assumptions

This thesis studies heterogeneous multi-teacher distillation for generative VLM students. The proposed method is not tied to a specific student parameter count. The experiments instantiate the setting with SmolVLM-500M and SmolVLM-256M because they provide practical student models for evaluating teacher-to-student transfer. The target setting is supervised visual question answering, where each example consists of an image, a natural-language question and a short answer sequence. The student is trained from labeled answer tokens and teacher output distributions, not from open-ended dialogue data or large-scale vision-language pretraining.

The method assumes white-box access to the teacher models during data preparation. In particular, teacher logits over supervised answer positions are available for distillation. The teachers are frozen throughout training and only the student model and the teacher routing network are updated. This assumption is important because the proposed trie-Wasserstein loss, the tokenizer-mismatched baselines and the gradient-agreement scores all depend on teacher-side probability distributions, not only final text responses.

The work does not assume that teachers and students share the same tokenizer, vocabulary, architecture or model family. This is the main reason for using tokenizer-agnostic distribution matching instead of direct token-index KL divergence. The teacher pool may contain models with different strengths and the method treats teacher usefulness as sample-dependent. GRACE therefore combines a tokenizer-agnostic KD loss with sparse routing and post-routing gradient refinement.

The empirical scope is limited to SmolVLM-family students and three VQA benchmarks: TextVQA, DocVQA and ChartQA. These datasets cover scene-text reading, document understanding and chart reasoning, but they do not exhaust all multimodal tasks. The method is also answer-token-centric: it distills teacher output distributions on supervised answer positions and does not explicitly align visual tokens, cross-attention maps or region-level evidence. These choices keep the training objective practical, but they also define the limits of the claims made in this thesis.

1.7 Thesis Organization

The remainder of the thesis is organized as follows. Chapter 2 introduces the technical background needed to understand GRACE. It begins with Transformer-based vision-language models and token-level knowledge distillation, then narrows to the specific difficulties considered in this thesis: tokenizer mismatch, Wasserstein and tree-structured transport, sparse teacher routing and gradient conflict.

Chapter 3 presents the proposed method. It first formulates heterogeneous teacher-to-student VLM distillation, then defines the main components of GRACE: the trie-Wasserstein KD loss, the teacher routing network, sparse top- k teacher selection and gradient-agreement refinement. The chapter ends by combining these components into the final optimization objective.

Chapter 4 describes the evaluation methodology. It introduces the three benchmark tasks, the student and teacher models, the training configuration, the evaluation metrics and the compared baselines. It also records implementation details needed for reproducibility, including offline teacher-logit preparation, remote logit storage and lazy streaming during training.

Chapter 5 reports the experimental results. It compares GRACE with direct fine-tuning, same-family logits KD, ULD, uniform multi-teacher averaging, hidden-state matching and reinforced teacher selection. It then analyzes representation similarity, component ablations, routing behavior and efficiency.

The final chapter summarizes the findings, discusses the main limitations and outlines future work, including repeated-seed evaluation, lower-cost teacher-logit storage and explicit visual-token or region-level distillation.

Chapter 2

Preliminaries

This chapter reviews the background needed to define GRACE. It covers VLM Transformers, knowledge distillation, tokenizer independent output matching, tree transport distances and sparse routing.

2.1 Transformer-Based Vision-Language Models

2.1.1 Transformer Basics

Modern vision-language models are built on the Transformer architecture [15]. For an input sequence, self-attention maps query, key and value matrices $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ to

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}, \quad (2.1)$$

where d_k is the key dimension. Multi-head attention runs this operation in parallel across several learned subspaces, then applies a feed-forward network and residual normalization.

For language modeling, the Transformer maps each hidden state $\mathbf{h}_j$ to next-token logits and probabilities:

$$\mathbf{z}_j = \mathbf{W}_o \mathbf{h}_j + \mathbf{b}_o, \quad p_j(v) = \text{softmax}(\mathbf{z}_j)_v = \frac{\exp(z_{j,v})}{\sum_{u \in \mathcal{V}} \exp(z_{j,u})}. \quad (2.2)$$

The proposed distillation method compares teacher and student distributions, such as $p_{t,j}$ and $p_{s,j}$, at supervised answer positions [15, 21].

2.1.2 Vision Encoder and Multimodal Fusion

On the visual side, many multimodal systems use a Vision Transformer (ViT) [45], which converts an image into a sequence of patch embeddings and processes them with Transformer blocks. A connector module then fuses these visual tokens with text, using cross-attention, a Q-Former, or a learned projector [3–5].

The distillation setting in this thesis assumes a generative VLM student that produces answer tokens autoregressively, which can be abstracted as a mapping from multimodal input to output logits:

$$\mathbf{z} = f_\theta(x),$$

where x is a multimodal input and $\mathbf{z}$ denotes the output logits used for autoregressive answer generation.

Figure 2.1 gives the two building blocks used in this abstraction: the Transformer attention block in Figure 2.1a and the ViT image-token pipeline in Figure 2.1b.

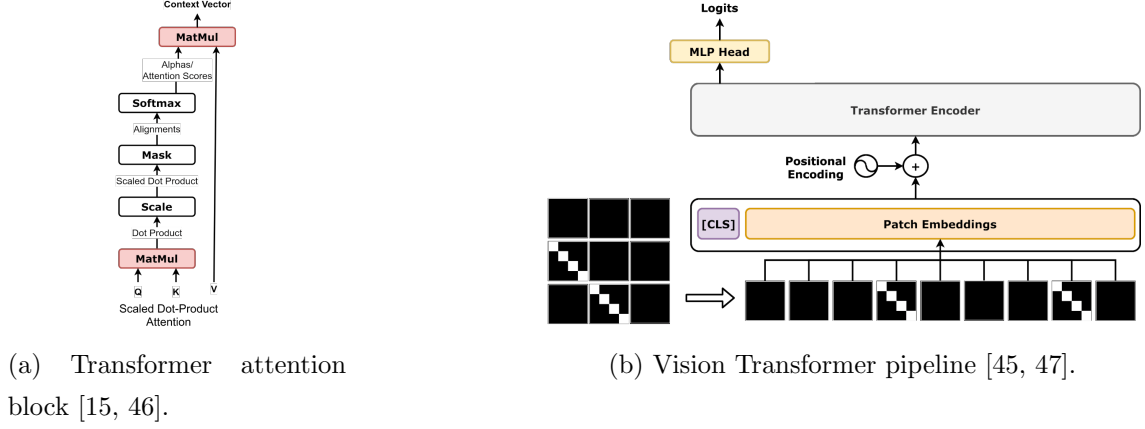


Figure 2.1: Transformer and ViT building blocks.

The experiments in this thesis focus on multimodal understanding, where the model must read image text, interpret the question and generate a short answer sequence [7–9]. Figure 2.2 summarizes representative architecture families of modern VLMs.

2.2 Knowledge Distillation

2.2.1 Teacher-Student Learning

Knowledge distillation transfers information from a stronger teacher model to a student model [21, 22]. Let $\mathcal{L}_{CE}$ denote the supervised cross-entropy loss of the student. A generic distillation objective takes the form

$$\mathcal{L} = \lambda_{CE}\mathcal{L}_{CE} + \lambda_{KD}\mathcal{L}_{KD}, \quad (2.3)$$

where $\mathcal{L}_{KD}$ is a teacher-guided loss. In later chapters, GRACE instantiates this template with a routed trie-Wasserstein term plus small router regularizers.

In token-level distillation, the teacher and student are compared on their output logits, not only on hard labels. The student therefore learns richer distributional structure, often called *dark knowledge* [21].

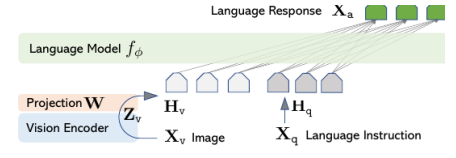
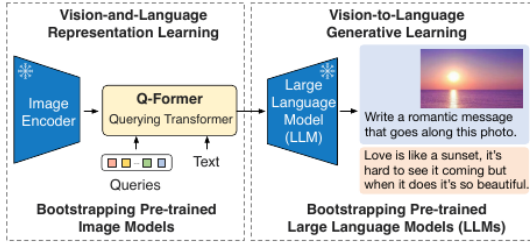
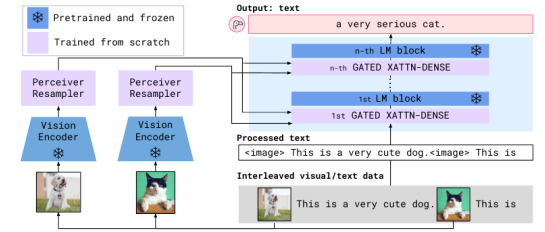
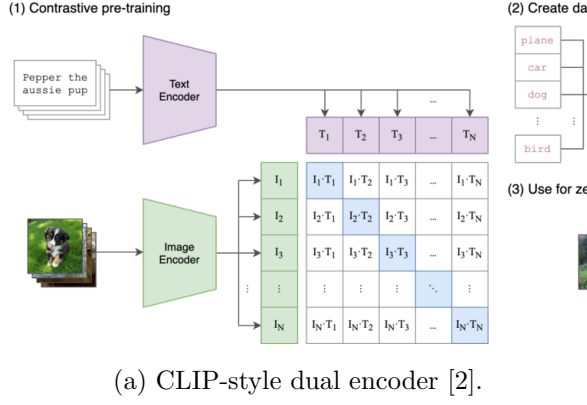


Figure 2.2: Representative architecture families of modern vision-language models.

2.2.2 Temperature-Scaled Output Distributions

Let $\mathbf{z}_s$ and $\mathbf{z}_t$ denote student and teacher logits. With temperatures τ_s and τ_t , the corresponding distributions are

$$\mathbf{p}_s = \text{softmax}\left(\frac{\mathbf{z}_s}{\tau_s}\right), \quad \mathbf{p}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau_t}\right). \quad (2.4)$$

Many classical KD methods compare $\mathbf{p}_s$ and $\mathbf{p}_t$ using divergences such as forward KL or Jensen-Shannon divergence. However, such approaches usually assume that teacher and student share the same vocabulary [21, 25, 48].

2.2.3 Answer-Token Distillation

For multimodal question answering, distillation should focus on *answer tokens*, not *prompt tokens* [5]. The KD loss is therefore applied only on supervised answer positions after masking ignored tokens, removing EOS when configured and truncating student and teacher answer spans to a common length.

In Chapter 3, these sets are denoted by Ω_b^{CE} for supervised CE positions and $\Omega_{b,i}^{\text{KD}}$ for teacher-specific KD positions. The result is a token-aligned distillation signal that remains compatible with different teacher and student tokenizers.

2.3 Heterogeneous Teacher Distribution Matching

2.3.1 ULD as a Prior Baseline

When the teacher and student vocabularies differ, KD losses defined on a shared output space are not directly applicable. Universal Logit Distillation (ULD) is a prior tokenizer-agnostic remedy because it avoids direct token-ID matching [25].

Given student logits $\mathbf{z}_s \in \mathbb{R}^{V_s}$ and teacher logits $\mathbf{z}_t \in \mathbb{R}^{V_t}$, we first compute their temperature-scaled probability distributions $\mathbf{p}_s$ and $\mathbf{p}_t$. We then sort both vectors in descending order, pad the smaller vocabulary with zeros and compare the sorted masses with an L_1 distance:

$$\mathcal{L}_{ULD}(\mathbf{z}_s, \mathbf{z}_t) = \|\pi(\mathbf{p}_s) - \pi(\mathbf{p}_t)\|_1, \quad (2.5)$$

where $\pi(\cdot)$ denotes sorting in descending order after zero-padding.

The key advantage of ULD is that it compares *ranked probability vectors* instead of token identities and is therefore directly applicable to teacher-student distillation with mismatched tokenizers [25]. In this thesis, ULD serves as the main tokenizer-mismatched baseline, but its ranked-mass matching procedure discards relations between decoded token strings. Two teacher-student token pairs with the same ranked probability shape receive the same loss even when one pair shares a long prefix and the other does not. This limitation motivates alignment methods that compare token distributions using a ground geometry over decoded token strings instead of only their ranked probability masses.

Figures 2.3 and 2.4 summarize the ULD baseline and the tokenizer-vocabulary mismatch that motivates tokenizer-agnostic distillation.

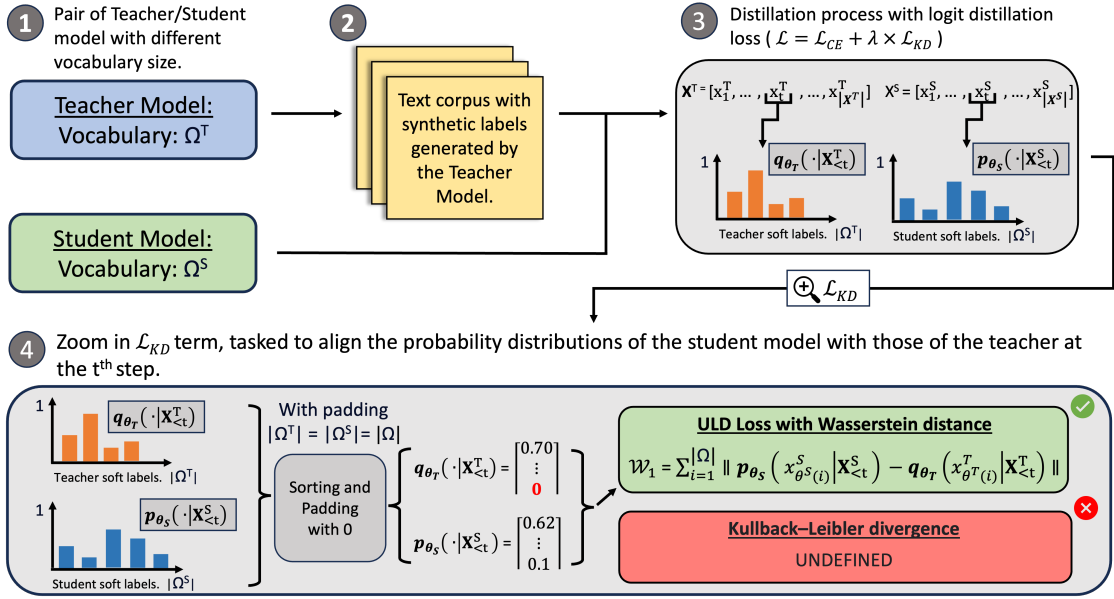


Figure 2.3: ULD distillation procedure. Teacher and student probability vectors are sorted in descending order, zero-padded to a common length and compared with an L_1 distance [25].

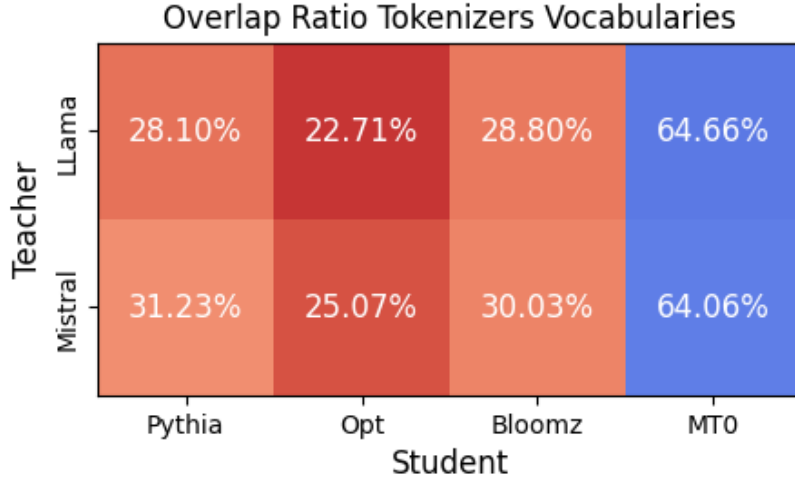


Figure 2.4: Tokenizer vocabulary overlap example. Low overlap across tokenizer vocabularies illustrates why direct token-ID matching is fragile when tokenizers differ.

2.3.2 Wasserstein Distance and Discrete Optimal Transport

The main weakness of rank-based comparison is that it does not encode the geometry of the output space. If two target distributions move farther away from a source distribution, a *pointwise divergence* may saturate once the supports no longer overlap. In contrast, Wasserstein distances continue to reflect the displacement cost between distributions.

Figure 2.5 illustrates this behavior. The source distribution is fixed, while the target distribution shifts progressively farther away. The Wasserstein curves grow with the displacement, whereas pointwise divergences such as total variation and squared Euclidean distance can saturate. Tokenizer-mismatched distillation benefits from this property because teacher and student probabilities may lie on mismatched token supports.

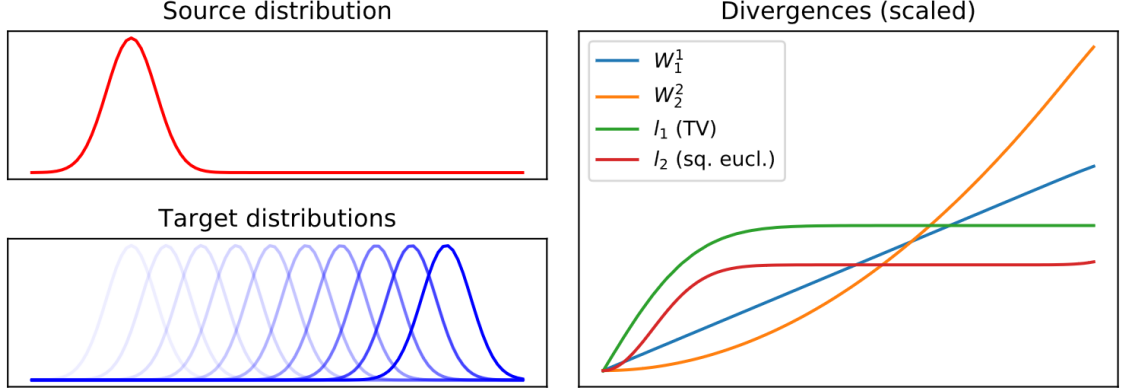


Figure 2.5: Geometric behavior of Wasserstein distance. The source distribution is compared with target distributions that shift progressively farther away. Pointwise divergences can saturate when supports stop overlapping, while Wasserstein distances continue to reflect displacement through the transport cost. Adapted from Flamary and Courty [49].

Optimal transport compares two probability measures by the minimum cost required to move mass from one distribution to the other [38]. For probability measures μ_s and μ_t , the p -Wasserstein cost can be written as

$$W_p^p(\mu_s, \mu_t) = \min_{\gamma \in \Pi(\mu_s, \mu_t)} \int_{\Omega_s \times \Omega_t} \|x - y\|^p d\gamma(x, y), \quad (2.6)$$

where $\Pi(\mu_s, \mu_t)$ is the set of couplings whose marginals are μ_s and μ_t . The coupling γ specifies how probability mass from the source distribution is transported to the target distribution.

For discrete source weights $\mathbf{a}$, target weights $\mathbf{b}$ and transport cost matrix $\mathbf{C}$, the corresponding optimization problem can be written as

$$W(\mathbf{a}, \mathbf{b}) = \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{P}, \mathbf{C} \rangle, \quad \mathcal{U}(\mathbf{a}, \mathbf{b}) = \{ \mathbf{P} \geq 0 \mid \mathbf{P}\mathbf{1} = \mathbf{a}, \mathbf{P}^\top \mathbf{1} = \mathbf{b} \}. \quad (2.7)$$

Here, $\mathbf{P}$ is a transport plan that specifies how much mass is moved from each source bin to each target bin.

2.3.3 Tree-Structured Mass Matching

Dense OT requires a pairwise ground cost between all source and target bins. In tokenizer-mismatched KD, however, token identity matching is not feasible because teacher and student vocabularies differ. A *tree metric* provides a structured alternative: probability mass is placed on a shared hierarchy and the distance is computed through *subtree-mass imbalance* [25, 38].

For W_1 on a tree metric, *tree-Wasserstein* distances make that idea explicit by placing probability mass on the tree and summing weighted subtree imbalances [26, 38]. For a rooted tree $T = (V, E)$ with edge weights $w(e)$, one convenient form is

$$W_T(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|, \quad (2.8)$$

where $\Gamma(e)$ denotes the subtree below edge e . The key point is that, on a tree metric, transport decomposes into edgewise subtree-mass imbalance instead of a dense source-target coupling [26, 37]. The computation is efficient because the comparison follows the tree hierarchy directly and the tree structure is a good fit when tokens share common byte prefixes or other hierarchical overlap.

Tree-sliced OT extends this idea by averaging Wasserstein distances over tree metrics and shows that tree metrics preserve the geometric view of OT while admitting fast closed-form computations [37]. In this work, the relevant tree is the shared UTF-8 byte trie constructed from decoded teacher and student single-token strings, not an abstract hierarchy.

This tree-structured view motivates the trie-Wasserstein loss introduced in Chapter 3: the method maps tokens to paths on a shared byte trie and compares mass on edges instead of token index.

Figures 2.6a and 2.6b show the two OT views used later: a tree-metric view for fast subtree-mass comparisons and a matrix view for discrete transport plans.

2.4 Multi-Teacher Distillation and Routing

2.4.1 Per-Teacher KD Loss

Let $\mathcal{T} = \{T_1, \dots, T_N\}$ be a pool of frozen teachers. For sample b , teacher T_i produces a per-teacher distillation loss $\mathcal{L}_{KD}^{(b,i)}$. In principle, one could average these losses uniformly,

$$\mathcal{L}_{KD}^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{KD}^{(b,i)}, \quad (2.9)$$

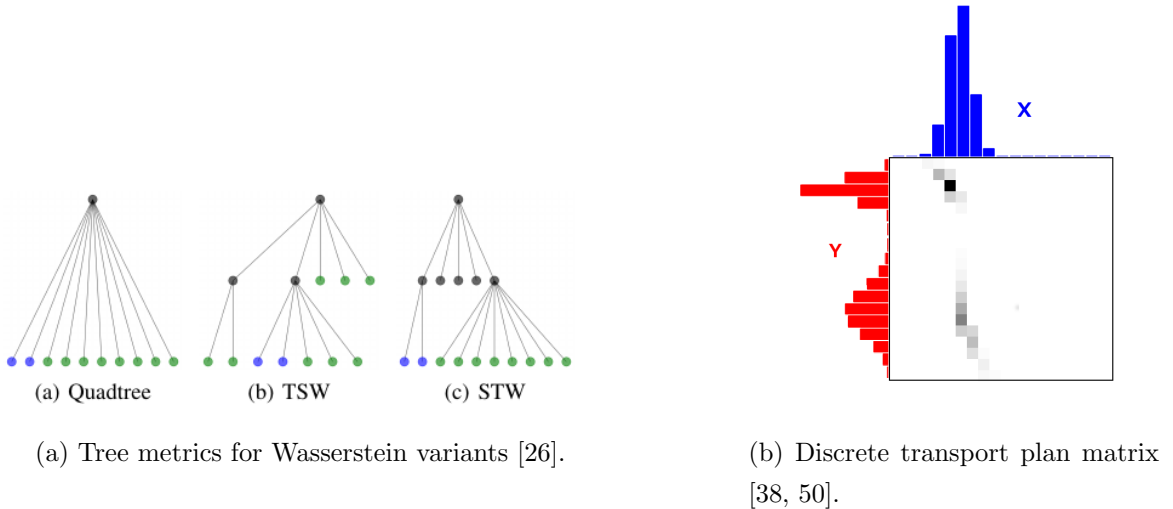


Figure 2.6: Tree-structured and matrix views of discrete optimal transport.

but this assumes every teacher is equally useful for every sample.

In practice, different teachers may provide conflicting guidance because their predictions, specialized strengths and KD gradients need not agree on the same training sample. Some teachers may help the supervised objective on a given sample, while others may push the student in a different direction [44]. This motivates a selective routing mechanism that decides, for each sample, which teachers should be active.

2.4.2 Learned Routing and Load Balancing

Recent sparse expert models route each sample or token through only a subset of available experts. This improves efficiency and allows specialized computation [41, 42, 51].

The same assignment principle can be applied to multi-teacher distillation by treating teachers as experts whose usefulness varies across samples. Instead of averaging all teachers equally, a learned router can propose teacher weights from student-side features [33, 34, 40].

Let $\mathbf{h}^{(b)}$ denote a pooled student representation for sample b . A router network $G_\phi(\cdot)$ produces gate logits

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}), \quad \tilde{\mathbf{r}}^{(b)} = \text{AdjustedScores}(\mathbf{r}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\tilde{\mathbf{r}}^{(b)}), \quad (2.10)$$

where $p_i^{(b)}$ is the initial routing weight for teacher T_i . The adjusted scores $\tilde{\mathbf{r}}^{(b)}$ may include expert bias, training noise, or temperature scaling before the routing softmax.

In sparse routing, these weights are subject to additional constraints that keep routing focused and prevent degenerate teacher assignments. *Top- k routing* retains only the highest-scoring teachers and zeros out all others. *Capacity constraints* prevent one teacher from receiving too many assignments. *Load balancing* prevents routing collapse onto a small subset of teachers.

Following the general MoE literature, one can summarize the balancing signal by a soft load $\bar{p}_i$ and a hard assignment load $\bar{q}_i$ after sparse routing, capacity control and fallback and define an auxiliary balancing objective of the form

$$\mathcal{L}_{bal} = N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (2.11)$$

This auxiliary term does not replace the KD objective. Instead, it regularizes the router so that different teachers remain usable during training.

Chapter 3 instantiates this principle with a learned MLP teacher router, top- k sparsification and routed-weight renormalization. Entropy and router z -loss terms regularize the gate. The current training configuration does not use capacity filtering or expert-bias updates.

The routing stage therefore resembles sparse-expert training more than static averaging of teacher losses [41–43].

The sparse-routing ideas that motivate the teacher gate in GRACE appear in Figures 2.7 and 2.8.

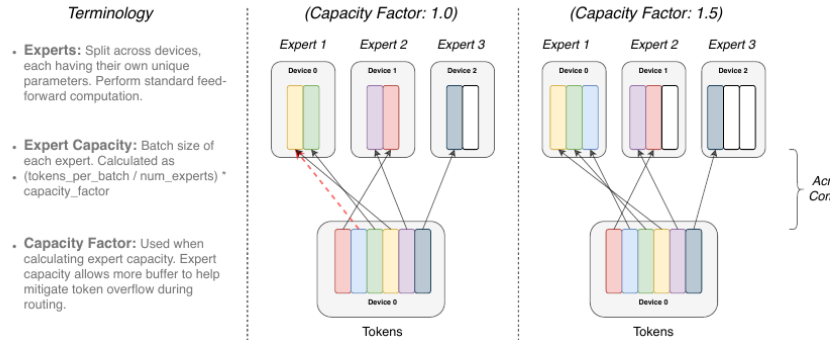


Figure 2.7: Sparse routing with expert capacity constraints, adapted from Switch Transformers [42].

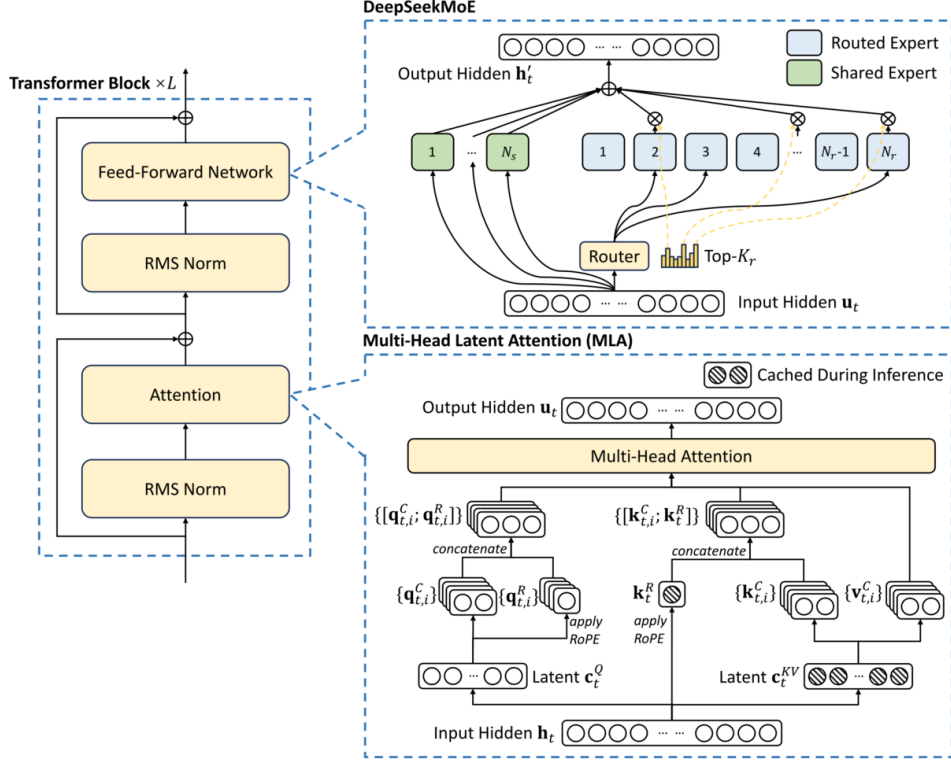


Figure 2.8: Sparse routing and expert specialization in a mixture-of-experts architecture [51, 52].

2.4.3 Gradient Conflict and Teacher Agreement

Routing alone does not fully solve multi-teacher conflict. Even after sparse routing selects a sparse teacher subset, the remaining teachers may still induce KD gradients that disagree with the supervised objective.

To see why, consider the standard combined distillation loss:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD}. \quad (2.12)$$

This objective minimizes a supervised term and a teacher-guided term together. Because both terms share the same student parameters, the combined update direction depends on whether the gradients $\nabla_{\theta} \mathcal{L}_{CE}$ and $\nabla_{\theta} \mathcal{L}_{KD}$ agree or conflict. When these two gradient directions are aligned, the combined update reinforces both objectives. When they oppose each other, the resulting update is a compromise that may degrade one or both objectives.

This is an instance of *multi-objective optimization*: a single parameter update must serve competing objectives simultaneously [44]. In multi-teacher distillation, the problem is compounded because each teacher contributes its own KD gradient. Let g_{CE} denote the supervised gradient of the student and let g_i^{KD} denote the KD gradient induced by teacher T_i . A teacher provides useful supervision when its KD gradient is broadly aligned with the supervised objective. It provides harmful supervision when the gradient points in an opposing direction. Prior work also uses gradient similarity in distillation as a signal for whether teacher supervision helps or conflicts with the student objective [44]. Agreement can be measured with *cosine similarity*:

$$s_i = \frac{\langle g_{\text{CE}}, g_i^{\text{KD}} \rangle}{\max(\|g_{\text{CE}}\|_2 \|g_i^{\text{KD}}\|_2, \epsilon_{\text{num}})}. \quad (2.13)$$

Here $\epsilon_{\text{num}} > 0$ is a numerical stabilizer. Positive values indicate agreement, negative values indicate conflict and values near zero indicate weak or nearly orthogonal alignment.

Figure 2.9 illustrates this intuition. Even when two objectives are individually meaningful, their local update directions can disagree, so a naive combined update may be suboptimal. This motivates post-routing refinement: after teacher selection, the final KD weights can be adjusted according to gradient agreement. Chapter 3 develops this idea into the GRACE gradient-agreement refinement mechanism.

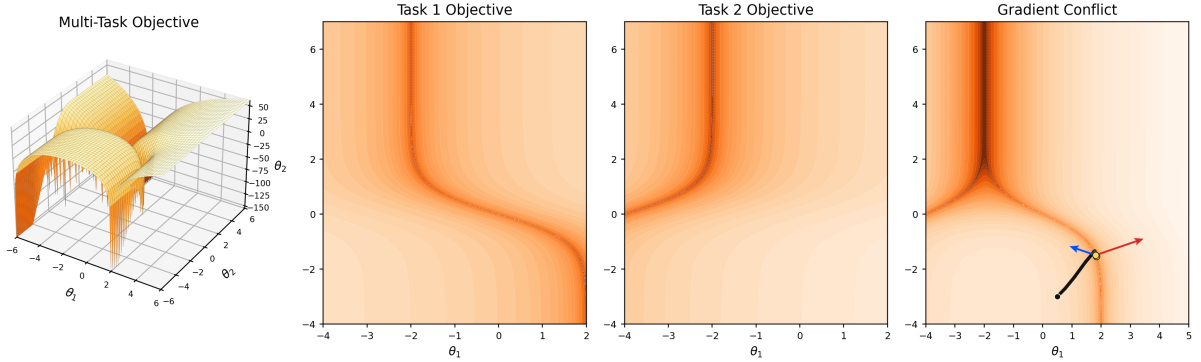


Figure 2.9: Gradient conflict in a multi-objective setting. The first three panels show the multi-task objective and two individual task objectives over the same parameter space. The final panel shows that different objectives can prefer different update directions. In multi-teacher distillation, an analogous conflict arises when a teacher KD gradient disagrees with the student’s supervised gradient.

Chapter 3

Proposed Method

This chapter presents **Gradient-Routed Alignment with Cross-tokenizer Experts (GRACE)** by first defining the heterogeneous multi-teacher distillation setting, then introducing the **trie-Wasserstein KD loss**, **teacher routing network**, **gradient-agreement refinement mechanism** and **final training objective**.

This chapter is organized as follows. Section 3.1 introduces the problem formulation and the overall GRACE training flow. Section 3.2 presents trie-Wasserstein distillation for mismatched tokenizers. Section 3.3 defines the teacher routing network and sparse top- k selection. Section 3.4 introduces gradient-agreement refinement and the final teacher weighting mechanism. Section 3.5 combines these components into the final optimization objective.

GRACE addresses three core difficulties in heterogeneous teacher-to-student VLM distillation. **First**, teacher and student models may use different tokenizers and output vocabularies, so standard token-index KD is not directly applicable. **Second**, teacher usefulness varies across image-question pairs, making fixed teacher averaging inefficient. **Third**, even routed teachers may still produce KD gradients that conflict with the supervised task objective. **Finally**, the method developed in this chapter addresses these issues through trie-Wasserstein KD, sparse teacher routing and gradient-agreement refinement.

3.1 Method Overview and Problem Formulation

We consider multimodal understanding tasks that require reading image text, understanding the question and generating an answer sequence [7–9]. The training set is

$$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}}, \quad (3.1)$$

where x_n is a multimodal input and y_n is the target answer sequence.

Let S_θ denote the student VLM with trainable parameters θ , let G_ϕ denote the trainable teacher router with parameters ϕ and let $\mathcal{T} = \{T_i\}_{i=1}^N$ be the frozen teacher pool. For each sample b , the student produces logits $\mathbf{z}_s^{(b)}$ and teacher T_i provides teacher logits $\mathbf{z}_t^{(b,i)}$. Throughout this chapter, the frozen teachers are treated as tokenizer-specific experts: each teacher may use a different tokenizer, vocabulary and model family and the router selects which experts contribute to each training sample.

GRACE avoids a fixed teacher subset by learning **sample-dependent sparse assignments** and refining the routed weights with **parameter-gradient agreement**.

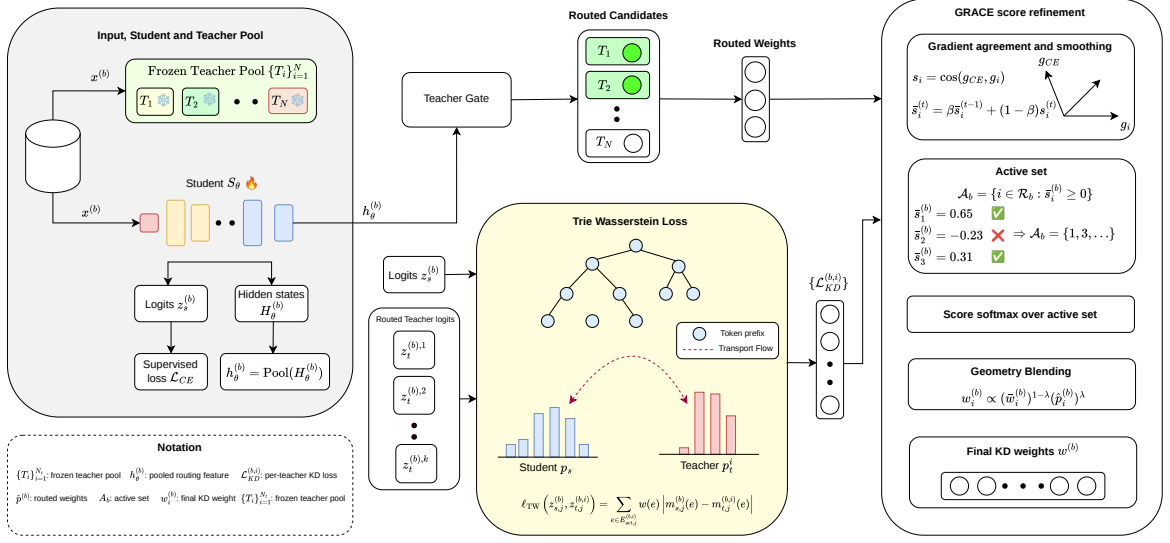


Figure 3.1: GRACE architecture. The student produces logits and hidden states, the teacher routing network selects candidate teachers, trie-Wasserstein KD computes per-teacher losses and gradient-agreement refinement produces final KD weights.

Figure 3.1 summarizes the full training path. The method is defined through four operators. **Trie-Wasserstein KD** computes per-teacher distillation losses on a shared byte trie. The **teacher routing network** selects candidate teachers from student-side hidden states. **Gradient-agreement refinement** adjusts routed teacher weights using parameter-gradient agreement. The **final objective** combines supervised CE, routed KD and router auxiliary terms.

GRACE adapts sparse expert routing [41, 42] to heterogeneous multi-teacher VLM distillation with token-level logit supervision [44].

Table 3.1 summarizes the main notation used in the method. Section 3.1.2 gives a concise overview of one GRACE training update and the following sections define these operators in detail, moving from tokenizer-agnostic distribution matching to teacher routing, gradient-agreement refinement and final loss aggregation.

3.1.1 Notation

Table 3.1: Main symbols used in Chapter 3.

Symbol	Meaning
$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}}$	Multimodal samples and target answer sequences.
$S_\theta, \mathcal{T} = \{T_i\}_{i=1}^N$	Trainable student VLM and frozen teacher pool.
G_ϕ	Trainable teacher routing network with parameters ϕ .
$\mathbf{z}_s^{(b)}, \mathbf{z}_t^{(b,i)}$	Student and teacher logits for sample b .
$\mathbf{H}_s, \mathbf{h}$	Student hidden states and pooled routing feature.
T_{train}	Total number of training steps.
$\rho_{\text{trie}}, K_{\text{TW}}, e_{\text{tail}}$	Trie edge decay, sparse support size and residual tail edge.
k_{route}	Configured router top- k .
$\mathbf{r}, \mathbf{p}$	Router logits and soft router weights.
$\mathcal{C}_b, \mathcal{R}_b, \hat{\mathbf{p}}^{(b)}$	Top- k candidate teacher set, routed teacher set and renormalized routed weights.
$s_i, \hat{s}_i, \tilde{a}_i, a_i^{(b)}, \mathcal{A}_b$	Raw agreement score, EMA-smoothed agreement score, teacher active flag, sample active flag and active routed set.
$\omega^{(b)}, \mathbf{w}^{(b)}$	Score-softmax weights and final KD weights.
$\Omega_b^{CE}, \Omega_{b,i}^{KD}$	Supervised CE positions and matched KD positions.
$\mathbf{g}_{CE}, \mathbf{g}_i$	CE and teacher-KD parameter gradients used by GRACE.
$\beta_{\text{EMA}}, \beta_{\text{score}}, \epsilon_{\text{num}}, \epsilon_{\text{fb}}, \lambda_{\text{blend}}$	EMA decay, score-softmax sharpness, cosine stabilizer, fallback threshold and router-score blend exponent.
TopKRoute	Sparse top- k teacher routing with routed-weight renormalization.
U	Uniform fallback weights over the active routed set.
Blend	Mixing routed weights with GRACE score weights.
EMA	Exponential moving average for GRACE scores.

Each GRACE update can be viewed as a four-stage training flow: **student forward pass**, **teacher routing**, **trie-Wasserstein KD computation** and **gradient-agreement refinement**. In the first stage, the student processes the mini-batch to produce answer-token logits $\mathbf{z}_s$, hidden states $\mathbf{H}_s$ and the supervised loss $\mathcal{L}_{CE}$. The router then computes gate logits $\mathbf{r}$, soft routing weights $\mathbf{p}$ and selects teachers with top- k masking. The third stage computes per-sample, per-teacher trie-Wasserstein losses $\mathcal{L}_{KD}^{(b,i)}$, which provide tokenizer-agnostic KD signals for the selected teachers. The final stage computes parameter-gradient agreement scores, smooths them over time, blends them with the routed weights and applies fallback when needed.

3.1.2 Training Flow

A GRACE update starts from a mini-batch $\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$ and a frozen teacher pool $\mathcal{T} = \{T_i\}_{i=1}^N$. The student S_θ first processes the multimodal inputs and returns answer-token logits $\mathbf{z}_s$, hidden states $\mathbf{H}_s$ and the supervised loss $\mathcal{L}_{CE}$. The supervised answer positions Ω_b^{CE} define both the cross-entropy loss and the pooling region used by the teacher router.

The router pools the student hidden states over supervised answer positions and maps the resulting feature $\mathbf{h}^{(b)}$ to teacher scores. After softmax normalization, top- k_{route} masking keeps a small routed teacher set $\mathcal{R}_b$ for each sample. The selected teacher weights are renormalized to obtain $\hat{\mathbf{p}}^{(b)}$. This stage replaces fixed teacher averaging with sample-dependent teacher selection.

For each routed teacher, GRACE compares the student answer-token distribution with the corresponding teacher distribution using trie-Wasserstein KD. In the implementation used for the experiments, teacher logits are generated offline and streamed during training, so the online update trains only the student and router while the teachers remain frozen. This produces a per-sample, per-teacher KD loss $\mathcal{L}_{KD}^{(b,i)}$.

During warmup, the final KD weights are the renormalized router weights. After warmup, GRACE computes a teacher-level agreement score by comparing the supervised CE gradient with each teacher’s batch-level KD gradient. The agreement scores are smoothed with EMA, intersected with the routed teacher set and converted into final sample-level KD weights. If no routed teacher passes the agreement threshold, GRACE falls back to the router weights. If active scores are nearly tied, it uses uniform active weights. Otherwise, it blends router weights with gradient-agreement weights.

The mini-batch KD loss is the weighted average of the routed per-teacher losses. The final objective combines supervised CE, routed trie-Wasserstein KD, router entropy regularization and router z-loss. The next sections define these components formally.

3.2 Trie-Wasserstein Cross-Tokenizer Distillation

3.2.1 Supervised Answer Tokens

We apply distillation only on **supervised answer tokens**, not on prompt tokens. For sample b , let Ω_b^{CE} denote the student positions whose labels are not ignored. For each teacher T_i , let $\Omega_{b,i}^{KD}$ denote the matched answer positions after masking ignored labels on both student and teacher sides, removing EOS from the student or teacher answer span when configured and truncating both spans to a common length.

This is a position-wise alignment approximation across tokenizers: after answer-span pre-processing, the j -th student and teacher answer positions are treated as corresponding positions. The trie-Wasserstein term handles token-distribution mismatch at each matched position, but it does not perform a separate byte-level sequence alignment.

3.2.2 Shared Byte Trie

For each sample b , token position $j \in \Omega_{b,i}^{KD}$ and teacher T_i , let $\mathbf{z}_{s,j}^{(b)} \in \mathbb{R}^{V_s}$ be the student logits and $\mathbf{z}_{t,j}^{(b,i)} \in \mathbb{R}^{V_i}$ be the teacher logits. The student and teacher probability distributions are

$$\mathbf{p}_{s,j}^{(b)} = \text{softmax}\left(\frac{\mathbf{z}_{s,j}^{(b)}}{\tau_s}\right), \quad \mathbf{p}_{t,j}^{(b,i)} = \text{softmax}\left(\frac{\mathbf{z}_{t,j}^{(b,i)}}{\tau_t}\right), \quad (3.2)$$

where τ_s and τ_t are the student and teacher temperatures.

Teacher and student vocabularies may differ, so direct token-index matching is not feasible. GRACE instead maps decoded teacher and student token strings into a **shared UTF-8 byte trie**. Tokens with common byte prefixes share the same initial trie path, even when their tokenizer IDs differ.

Using UTF-8 bytes is a necessary design choice because it provides a fixed alphabet and exposes tokenizer artifacts such as leading spaces, wordpiece prefixes and multimodal special tokens. Character-level matching is less stable because the same visible text can be stored with different Unicode code-point sequences.

For example, a string with a **leading space maps** to byte sequence $[32, 99, 97, 116]$, while **cat** maps to $[99, 97, 116]$. The trie therefore treats the leading-space byte as an **explicit edge**, while **preserving the subsequent byte sequence** for the underlying word.

Each token string is converted to a byte sequence and terminated with an EOS sentinel. The EOS marker separates complete tokens from strict prefixes, so tokens such as **doc** and **document** remain distinct leaves even though they share an initial path.

Let e denote a normal byte-trie or EOS edge and let $d(e)$ be its depth. GRACE assigns edge weights

$$w(e) = \rho_{\text{trie}}^{d(e)-1}, \quad (3.3)$$

where $0 < \rho_{\text{trie}} < 1$. The first edge from the root has weight 1 and deeper edges decay geometrically. This makes shared prefixes cheaper than unrelated byte paths while preserving a valid tree metric [26, 38].

Different tokenizers may stop at different depths, but they still share the same byte-prefix edges when their decoded token strings overlap in byte-prefix structure. The example in Figure 3.2 uses the pair `document/receipt` and `doc/rece`.

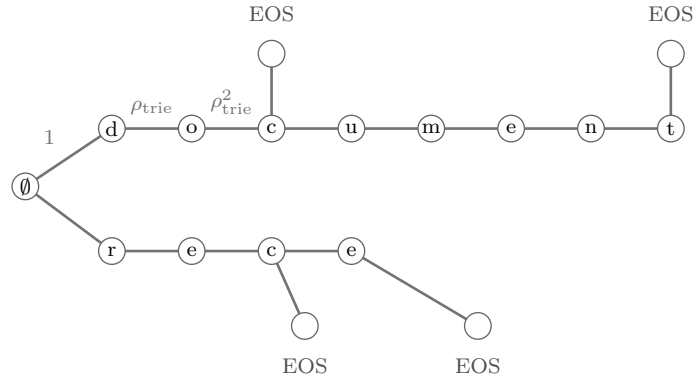


Figure 3.2: Shared-trie view of tokenizer-mismatched matching for `document/receipt` and `doc/rece`.

Figure 3.2 illustrates this effect: shorter teacher token sequences terminate earlier at EOS, whereas longer student token sequences extend farther down the trie. As a result, the mismatch is represented through the trie hierarchy instead of direct token-ID comparison.

3.2.3 Trie-Wasserstein Objective

The shared byte trie induces a tree metric over decoded token strings. Let $\mathcal{V}_s$ and $\mathcal{V}_t$ denote the student and teacher token vocabularies being compared at a given answer position and let $\pi(v)$ denote the trie path of decoded token v . For an edge e , the student and teacher subtree masses are

$$m_s(e) = \sum_{\substack{v \in \mathcal{V}_s \\ e \in \pi(v)}} p_s(v), \quad m_t(e) = \sum_{\substack{u \in \mathcal{V}_t \\ e \in \pi(u)}} p_t(u).$$

Thus, $m_s(e)$ and $m_t(e)$ measure how much probability mass each distribution assigns to token paths passing through edge e . The W_1 optimal transport distance between teacher and student token distributions on the trie can then be written as a weighted sum of subtree-mass imbalances:

$$W_T(p_s, p_t) = \sum_{e \in E} w(e) |m_s(e) - m_t(e)|. \quad (3.4)$$

This result follows from the standard W_1 Kantorovich optimal transport formulation under the trie-induced tree metric [26, 38]. The sparse construction below applies the same formula to a bounded $\text{top-}K_{\text{TW}}$ support with a residual tail edge.

3.2.4 Sparse Support and Residual Tail Edge

Equation (3.4) is exact on the complete shared trie, but evaluating all vocabulary tokens at every answer position is expensive. GRACE therefore applies the same subtree-mass formula on a **sparse augmented trie**. For each aligned answer position, the method first masks ignored vocabulary entries and retains the **top- K_{TW} student and teacher tokens**. These retained tokens are inserted into the shared UTF-8 byte trie while preserving their exact trie paths and shared prefix structure. The probability mass assigned to all remaining non-top- K_{TW} tokens is not discarded. Instead, it is collapsed into a single **residual tail edge**. The trie-Wasserstein loss is then computed on the augmented structure containing the retained token paths and the residual tail edge. This construction is exact for the collapsed sparse trie, but it approximates the complete trie distance because all omitted vocabulary mass is represented by one tail atom.

Let K_{TW} denote the sparse trie support size. Let $\mathcal{V}_s^*$ and $\mathcal{V}_t^*$ denote the non-ignored student and teacher vocabularies. Before constructing the sparse trie, the student and teacher probabilities are renormalized over the non-ignored vocabularies so that the retained and residual masses form valid distributions. Let $\mathcal{K}_{s,j}^{(b)} \subset \mathcal{V}_s^*$ and $\mathcal{K}_{t,j}^{(b,i)} \subset \mathcal{V}_t^*$ be the retained token sets. The residual mass assigned to the tail edge is

$$r_{s,j}^{(b)} = 1 - \sum_{v \in \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v), \quad r_{t,j}^{(b,i)} = 1 - \sum_{v \in \mathcal{K}_{t,j}^{(b,i)}} p_{t,j}^{(b,i)}(v). \quad (3.5)$$

Let $\pi(v)$ be the trie path for token v and let e_{tail} be the global residual tail edge with $w(e_{\text{tail}}) = 1$. For any active trie edge e , the sparse subtree masses are

$$m_{s,j}^{(b)}(e) = \sum_{\substack{v \in \mathcal{K}_{s,j}^{(b)} \\ e \in \pi(v)}} p_{s,j}^{(b)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{s,j}^{(b)}, \quad (3.6)$$

$$m_{t,j}^{(b,i)}(e) = \sum_{\substack{v \in \mathcal{K}_{t,j}^{(b,i)} \\ e \in \pi(v)}} p_{t,j}^{(b,i)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{t,j}^{(b,i)}. \quad (3.7)$$

The token-level trie-Wasserstein loss is then

$$\ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) = \sum_{e \in \mathcal{E}_{act,j}^{(b,i)}} w(e) \left| m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e) \right|, \quad (3.8)$$

where $\mathcal{E}_{act,j}^{(b,i)}$ contains the active student edges, active teacher edges and the residual tail edge. Before the absolute value is applied, $m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e)$ is positive when edge e carries more student mass and negative when it carries more teacher mass.

The per-sample, per-teacher KD loss is the mean over supervised answer tokens:

$$\mathcal{L}_{KD}^{(b,i)} = \frac{1}{|\Omega_{b,i}^{KD}|} \sum_{j \in \Omega_{b,i}^{KD}} \ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}). \quad (3.9)$$

The tail edge preserves probability mass outside the top- K_{TW} support without materializing every vocabulary path.

Complexity analysis. The shared byte trie is built once with time and memory cost $O(V_s \bar{L}_s + V_t \bar{L}_t)$, where V_s and V_t are the student and teacher vocabulary sizes and $\bar{L}_s$ and $\bar{L}_t$ are the average byte-path lengths. For one aligned answer position, extracting the top- K_{TW} student and teacher tokens costs $O(V_s \log K_{TW} + V_t \log K_{TW})$ with heap-based selection. Given the retained token sets, sparse subtree-mass accumulation walks only the retained token paths and the residual tail edge, costing $O(K_{TW} \bar{L}_s + K_{TW} \bar{L}_t + A)$, where A is the number of active trie edges. If the implementation explicitly sorts or deduplicates active edge indices, this adds an $O(A \log A)$ term. Otherwise, indexed accumulation gives an $O(A)$ edge-aggregation cost.

Runtime memory is $O(E + P_s + P_t + A)$, where E is the number of trie edges and P_s and P_t are the stored student and teacher path-index arrays. Unlike **dense OT**, the sparse computation does not materialize a $|V_s| \times |V_t|$ transport matrix.

Compared with the ULD baseline from Chapter 2, this loss compares **mass on shared trie edges** instead of **ranked logits**. It keeps the tokenizer-agnostic goal, but it uses a structured transport view that is closer to the actual tokenizer geometry, while the top- K_{TW} +tail approximation keeps the cost bounded by the retained support.

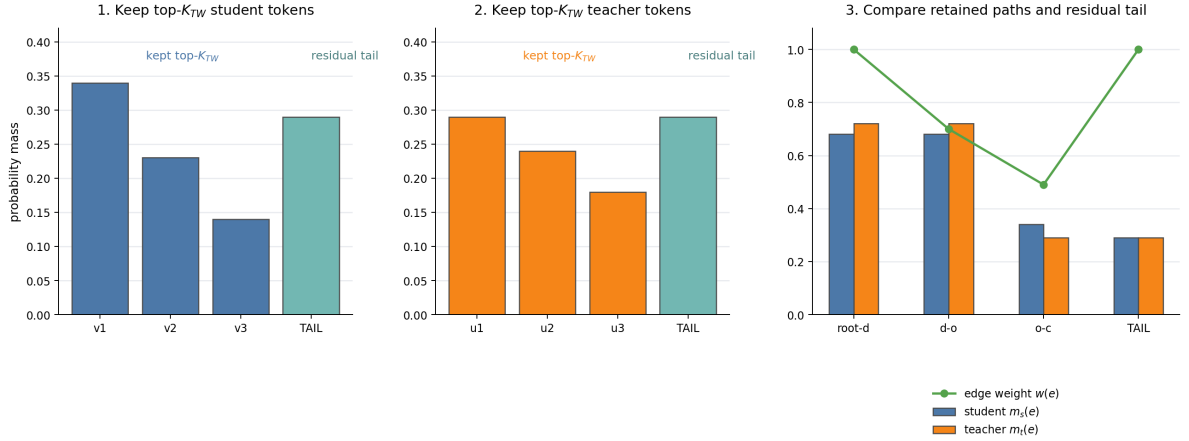


Figure 3.3: Sparse trie-Wasserstein support and residual tail mass.

Figure 3.3 shows how the left and middle panels keep top- K_{TW} student and teacher tokens while assigning the remaining probability to a residual tail edge. The right panel then compares signed subtree masses on retained trie edges and the tail edge.

Table 3.2 isolates the effect of **string geometry**. Both rows have identical ranked masses, so **ULD gives the same value**. Trie-Wasserstein assigns a lower loss to **doc/document** than to **abc/def** because the former pair shares a **byte prefix**.

Table 3.2: Illustrative comparison between ULD and trie-Wasserstein under the same ranked masses. ULD assigns the same loss to both rows, while trie-Wasserstein gives lower loss to the pair with a shared byte prefix.

Teacher / student top token	$\mathcal{L}_{ULD}$	ℓ_{TW}
doc / document	0.20	0.60
abc / def	0.20	3.29

3.3 Teacher Routing Network

3.3.1 Router Architecture

When multiple teachers are available, the method first builds a **teacher routing network** instead of averaging them uniformly. The router is applied to projector-based VLM students whose language-model head exposes the text hidden width.

Let $\mathbf{H}_s^{(b)}$ denote the student hidden states captured immediately before the language-model head for sample b . We pool these hidden states over supervised answer positions to obtain a **routing feature**

$$\mathbf{h}^{(b)} = \text{Pool}(\mathbf{H}_s^{(b)}) . \quad (3.10)$$

The pooling operation takes the mean over supervised answer positions. The routing feature is defined for samples with at least one supervised answer token, as ensured by the training data construction.

A learned MLP router $G_\phi(\cdot)$ then produces gate logits

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}) . \quad (3.11)$$

Soft routing weights are obtained by applying softmax to the router logits:

$$\mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)}) , \quad (3.12)$$

where $p_i^{(b)}$ is the initial gate weight for teacher T_i . The gate stack uses LayerNorm, an up projection, a gate-value split, SiLU gating and a down projection. The router then masks to the top- k_{route} teachers and renormalizes the selected softmax weights.

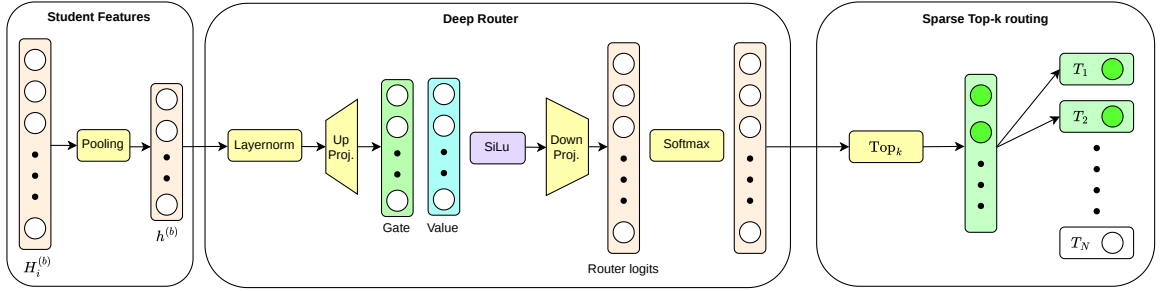


Figure 3.4: Teacher routing network. Student hidden states are pooled over supervised answer positions, passed through a gated MLP router, converted into soft teacher weights and sparsified by top- k routing.

Figure 3.4 expands the teacher router used in the training flow, showing the pooled student hidden state, gated MLP transformation, softmax teacher weights and top- k sparsification step.

3.3.2 Top-k Routing

After computing soft routing weights, GRACE converts them into a **sparse routed set**. For sample b , GRACE first selects the highest-scoring teachers according to the router logits:

$$\mathcal{C}_b = \text{TopK}(\mathbf{r}^{(b)}, \min(k_{\text{route}}, N)) . \quad (3.13)$$

The routed set is then defined as $\mathcal{R}_b = \mathcal{C}_b$. All non-selected teacher weights are set to zero and the remaining softmax weights are renormalized. Let $m_i^{(b)} = \mathbf{1}[i \in \mathcal{R}_b]$ denote the top- k assignment mask. The routed gate weights are

$$\hat{p}_i^{(b)} = \frac{m_i^{(b)} p_i^{(b)}}{\sum_{j=1}^N m_j^{(b)} p_j^{(b)}}. \quad (3.14)$$

The result is a **sparse routed teacher set** $\mathcal{R}_b$ and normalized **routed weights** $\hat{\mathbf{p}}^{(b)}$, guaranteeing that at least one teacher remains available before gradient-agreement refinement.

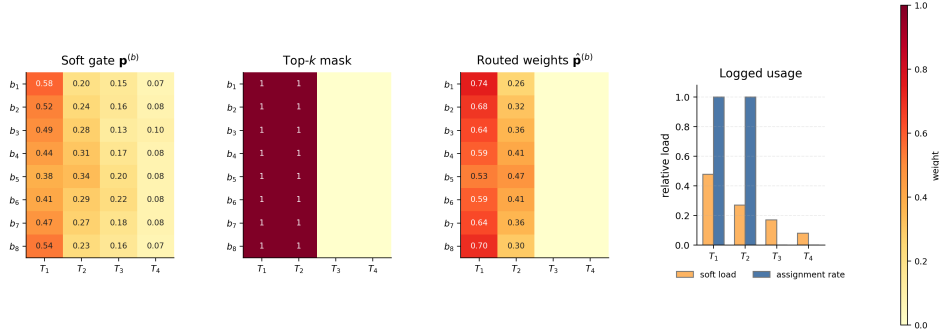


Figure 3.5: Top- k routing dynamics in GRACE.

Figure 3.5 contrasts the dense gate and routed assignment view: the dense gate assigns nonzero probability to every teacher, while top- k routing retains only the strongest candidates for each sample. The logged assignment rate summarizes how often each teacher appears in the routed set.

3.3.3 Router Entropy and Z-Loss

GRACE uses two auxiliary router terms to regularize the teacher-selection mechanism. The first is **entropy regularization**, implemented as a negative-entropy term that discourages premature routing collapse:

$$\mathcal{L}_{ent} = -\frac{1}{B} \sum_{b=1}^B H(\mathbf{p}^{(b)}). \quad (3.15)$$

The second is the **router z -loss**, which keeps router logits numerically well behaved:

$$\mathcal{L}_z = \frac{1}{B} \sum_{b=1}^B \left(\log \sum_{i=1}^N e^{r_i^{(b)}} \right)^2. \quad (3.16)$$

Neither term replaces the supervised CE or KD losses. They serve only to stabilize routing behavior during training.

3.4 Gradient-Agreement Refinement

As discussed in Section 2.4.3, routed teachers may still conflict with the supervised objective after teacher selection. **Gradient-agreement refinement** addresses this issue by checking whether each selected teacher’s KD direction agrees with the supervised CE direction.

3.4.1 Gradient Agreement Scoring

Let $\Theta \subseteq \theta$ denote the trainable student parameters used by the optimizer. GRACE scores each teacher by comparing the **supervised CE gradient** with each **teacher-specific KD gradient** in student-parameter space. The CE gradient is

$$\mathbf{g}_{CE} = \nabla_{\Theta} \mathcal{L}_{CE}. \quad (3.17)$$

For teacher T_i , GRACE computes the KD gradient from that teacher’s mean KD loss over the full mini-batch:

$$\mathbf{g}_i = \nabla_{\Theta} \left(\frac{1}{B} \sum_{b=1}^B \mathcal{L}_{KD}^{(b,i)} \right). \quad (3.18)$$

In this thesis, $\mathcal{L}_{KD}^{(b,i)}$ is the trie-Wasserstein loss from Equation 3.9. These gradients are used only to score teacher compatibility. The final parameter update still uses the full loss in Section 3.5.

3.4.2 Agreement Scores and Active Set

For each teacher T_i , GRACE measures directional agreement between supervised learning and KD by cosine similarity:

$$s_i = \frac{\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle}{\max(\|\mathbf{g}_{CE}\|_2 \|\mathbf{g}_i\|_2, \epsilon_{\text{num}})}. \quad (3.19)$$

The agreement score s_i is **teacher-level** within the current mini-batch. The same teacher score is broadcast across samples. Sample dependence enters later through the routed set $\mathcal{R}_b$, routed router weights $\hat{\mathbf{p}}^{(b)}$ and active set $\mathcal{A}_b$.

GRACE forms the active set by combining teacher-level agreement with sample-specific routing. The teacher-level agreement score is first converted into a binary teacher flag:

$$\tilde{a}_i = \mathbf{1}[s_i > \delta], \quad (3.20)$$

where δ is the GRACE threshold. This flag is then intersected with the routed teacher set for each sample:

$$a_i^{(b)} = \mathbf{1}[i \in \mathcal{R}_b] \tilde{a}_i, \quad \mathcal{A}_b = \{i \in \mathcal{R}_b : a_i^{(b)} = 1\}. \quad (3.21)$$

Thus, s_i is **teacher-level** within the mini-batch, while $\mathcal{A}_b$ remains **sample-specific** through $\mathcal{R}_b$. GRACE intentionally thresholds the current raw agreement score. The score-softmax and fallback spread below use the EMA-smoothed score.

The agreement scores are used as non-differentiable routing statistics when constructing the final KD weights. The final optimization step does not backpropagate through the gradient-agreement computation itself.

3.4.3 Warmup, EMA Smoothing and Score Weights

The refinement stage uses **warmup**, **EMA smoothing** and **score-softmax weighting** to stabilize teacher weights. Let $\rho_g \in [0, 1]$ denote the GRACE warmup ratio relative to the total number of training steps. During the first $\rho_g T_{\text{train}}$ steps, GRACE leaves the routed gate weights unchanged:

$$w_i^{(b)} = \hat{p}_i^{(b)}. \quad (3.22)$$

After warmup, GRACE refines the routed teachers with a smoothed score distribution. Let $\bar{s}_i^{\text{prev}}$ be the running exponential moving average (EMA) of past batch means for teacher T_i and let β_{EMA} be the EMA decay. The smoothed score is

$$\hat{s}_i = \beta_{\text{EMA}} \bar{s}_i^{\text{prev}} + (1 - \beta_{\text{EMA}}) s_i. \quad (3.23)$$

Smoothed scores produce more stable teacher weights, consistent with prior findings on variance reduction in stochastic optimization. If $\mathcal{A}_b = \emptyset$, GRACE keeps the routed weights $\hat{\mathbf{p}}^{(b)}$. Otherwise, it converts smoothed scores into a score-softmax weight vector over $\mathcal{A}_b$:

$$\omega_i^{(b)} = \frac{\exp(\beta_{\text{score}} \hat{s}_i) \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \exp(\beta_{\text{score}} \hat{s}_j) \mathbf{1}[j \in \mathcal{A}_b]}, \quad (3.24)$$

Here β_{score} controls how sharply the highest-scoring active teacher is preferred.

3.4.4 Weight Blending and Fallback

GRACE converts routed teacher scores into final KD weights through three mutually exclusive cases.

No active teacher. If $\mathcal{A}_b = \emptyset$, GRACE retains the routed weights:

$$w_i^{(b)} = \hat{p}_i^{(b)}. \quad (3.25)$$

Low-confidence scores. If active teachers exist but their smoothed scores are too close, GRACE uses uniform weights over the active set. The score range is

$$\Delta^{(b)} = \max_{i \in \mathcal{A}_b} \hat{s}_i - \min_{i \in \mathcal{A}_b} \hat{s}_i. \quad (3.26)$$

If $\mathcal{A}_b \neq \emptyset$ and $\Delta^{(b)} < \epsilon_{fb}$, then

$$w_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{A}_b]}{|\mathcal{A}_b|}. \quad (3.27)$$

Confident scores. Otherwise, GRACE blends router weights with score weights. Let $\lambda_{\text{blend}} \in [0, 1]$ be the router-blend exponent. The unnormalized blended weights are

$$\tilde{w}_i^{(b)} \propto \left(\hat{p}_i^{(b)}\right)^{\lambda_{\text{blend}}} \left(\omega_i^{(b)}\right)^{1-\lambda_{\text{blend}}}, \quad (3.28)$$

followed by normalization over the active routed teachers:

$$w_i^{(b)} = \frac{\tilde{w}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \tilde{w}_j^{(b)} \mathbf{1}[j \in \mathcal{A}_b]}. \quad (3.29)$$

Equations 3.24, 3.27 and 3.29 define the gradient-refinement operator. The router first chooses $\mathcal{R}_b$, GRACE filters it into the active set $\mathcal{A}_b$ and the final weights $w_i^{(b)}$ are either **inherited from the router**, **assigned uniformly under low score spread**, or **obtained by a geometric blend** of $\hat{p}_i^{(b)}$ and $\omega_i^{(b)}$. Figure 3.6 shows these weighting regimes.

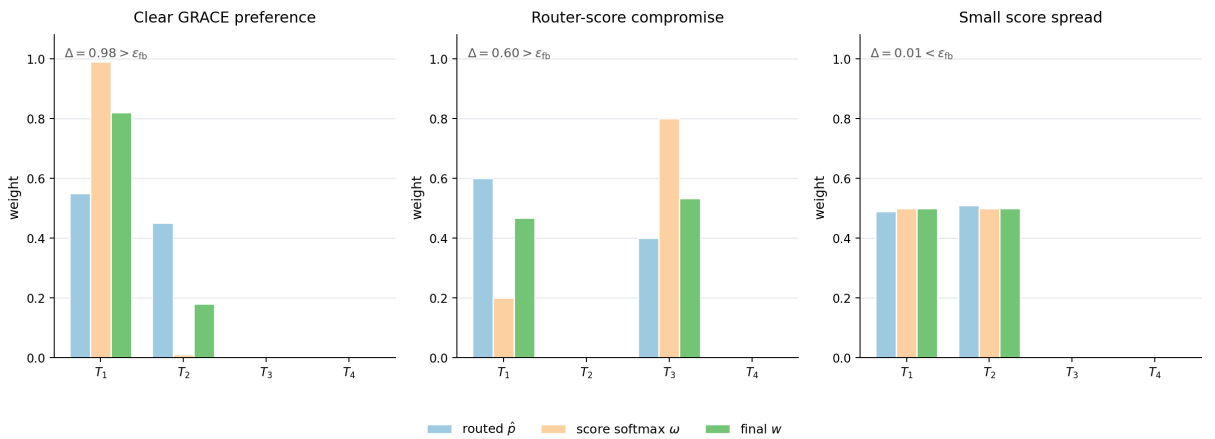


Figure 3.6: Representative KD weighting regimes.

In all cases, GRACE only refines weights within the routed set, except when no teacher passes the active threshold.

3.4.5 Per-Sample KD Aggregation

The **final KD loss for sample b** is the weighted sum of routed per-teacher losses:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (3.30)$$

The **mini-batch KD term** is

$$\mathcal{L}_{KD} = \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)}. \quad (3.31)$$

3.5 Final Objective

The preceding sections define the three components needed for the full GRACE objective. First, **supervised CE** anchors the student to ground-truth answer tokens. Second, **routed KD** transfers teacher supervision after sparse routing and gradient-agreement refinement. Third, **router regularization** stabilizes the teacher gate during training.

The final optimization objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD} \mathcal{L}_{KD} + \lambda_{ent} \mathcal{L}_{ent} + \lambda_z \mathcal{L}_z, \quad (3.32)$$

Here, α_{KD} controls the strength of teacher supervision relative to the supervised answer-token loss. The coefficient λ_{ent} weights the entropy regularizer, which discourages premature collapse of the teacher gate, while λ_z weights the router z -loss, which keeps the gate logits numerically stable.

The objective is minimized with respect to (θ, ϕ) , where θ are the student parameters and ϕ are the router parameters. All teacher parameters remain frozen throughout training. In the experimental implementation, teacher logits are generated offline and streamed during training, so the online update does not require backpropagation through the teacher pool.

During warmup, the KD weights are the renormalized top- k router weights. After warmup, GRACE computes teacher-level gradient agreement, smooths the scores with EMA and refines the KD weights within each routed teacher set. Thus, routing acts as a candidate-selection mechanism, while gradient agreement acts as a corrective weighting mechanism.

Equations (3.30) through (3.32) specify how per-teacher losses are converted into the training loss used in the experiments. The resulting objective addresses the three main difficulties of heterogeneous multi-teacher distillation: trie-Wasserstein KD handles tokenizer mismatch, sparse routing handles sample-dependent teacher utility and gradient-agreement refinement suppresses routed teachers whose KD gradients conflict with supervised learning.

Chapter 4

Evaluation Methodology

4.1 Experimental Setup

All experiments follow a shared evaluation protocol for fair comparison across methods.

The experiments instantiate the student side with **SmolVLM-500M** and **SmolVLM-256M**, evaluated on multimodal benchmarks [16].

All models are evaluated with **VLM-Eval-Kit** [53], which provides a shared inference, post-processing and metric-computation pipeline for the three benchmarks. In the experiments, the toolkit standardizes evaluation by loading each benchmark split, applying the appropriate input template, running inference through a unified model wrapper, normalizing the generated answers and computing the dataset-specific score.

4.2 Datasets

We evaluate three text-centric VQA benchmarks: **TextVQA**, **DocVQA** and **ChartQA**.

These benchmarks test whether the model can **read visual text, understand questions and perform basic reasoning** across natural images, documents and charts. They were chosen because they cover complementary visual domains and combine OCR-heavy understanding with layout interpretation and numerical reasoning [7–9].

4.2.1 Dataset Descriptions

TextVQA [7] evaluates question answering on natural images containing scene text. It focuses on reading scene text in natural images and combining it with question understanding. *Sample question*: “What word is written on the stop sign?” *Sample answer*: “No word is visible on this sign.”

DocVQA [8]: evaluates VQA on real document images (scans/photos), where layout, paragraphs, tables and forms are important. It emphasizes document structure understanding in addition to OCR. *Sample question*: “What is the invoice number shown in this document?” *Sample answer*: “4572.”

ChartQA [9]: evaluates question answering over charts. It tests chart reading together with numerical and logical reasoning to derive the final answer. *Sample question*: “Which year has the highest sales value in the chart?” *Sample answer*: “The second (middle) bar has the highest value.”



Figure 4.1: Representative examples from the three evaluation domains [54–56].

The evaluation therefore covers natural scene text, document layouts and chart reasoning, as illustrated in Figure 4.1.

4.2.2 Dataset Size and Splits

Table 4.1 lists the split sizes used throughout the evaluation chapter.

Table 4.1: Train, validation and test split sizes for TextVQA, DocVQA and ChartQA [7–9]. ChartQA is also broken down into its human and machine-generated subsets.

Dataset	Train	Val	Test	Total
TextVQA	34,602	5,000	5,734	45,336
DocVQA	39,463	5,349	5,188	50,000
ChartQA-H (human)	7,398	960	1,250	9,608
ChartQA-M (machine)	20,901	960	1,250	23,111
ChartQA (H+M, combined)	28,299	1,920	2,500	32,719

4.3 Training Configuration

Unless otherwise stated, the main distillation experiments use a shared training configuration so that differences in performance can be attributed to the distillation strategy, not unrelated optimization choices.

We run all training jobs on a single high-memory **NVIDIA H200** graphics processing unit (GPU) on **Modal** or on comparable single-GPU **Vast.ai** instances, depending on availability. A typical full H200 run takes about 7 hours and costs about 14.41 USD at 2.059 USD/hour on Vast.ai, excluding the separate cost of offline teacher-logit generation and remote storage. Distillation training uses **DeepSpeed ZeRO Stage 2**, **Brain Floating Point 16 (bf16)** mixed precision, **TensorFloat-32 (TF32)** matrix multiplication and gradient checkpointing to control memory use while keeping the student fully trainable and all teachers frozen [57, 58].

Before training, we convert all benchmark samples into a unified **LLaVA-style multi-modal instruction format** [5]. Each example is stored as an image plus a short user-assistant dialogue, so TextVQA, DocVQA and ChartQA can share the same data loader, prompt template and answer supervision interface.

Listing 4.1 shows the structure used in our preprocessing pipeline.

```

1 {
2   "image": "sample_000123.jpg",
3   "conversations": [
4     {"from": "human", "value": "<image>\nWhat is the invoice number?"},
5     {"from": "gpt", "value": "4572"}
6   ]
7 }
```

Listing 4.1: Representative LLaVA-style sample used after dataset conversion.

Table 4.2 summarizes the main optimization and routing settings used in the main training configuration. The engineering details of teacher-logit storage and remote caching are deferred to Section 4.6, because they support reproducibility but are not part of the fair-comparison protocol itself.

Table 4.2: Main optimization and routing settings for the active DocVQA training configuration. These values define the fair-comparison protocol used for the main training recipe, while broader engineering details are deferred to Section 4.6.

Setting	Value
Student model	HuggingFaceTB/SmolVLM-500M-Instruct
	Qwen/Qwen2.5-VL-3B-Instruct
Teacher models	Qwen/Qwen2-VL-2B-Instruct
	OpenGVLab/InternVL3-1B
	google/gemma-3-4b-it

Continued on the next page.

Table 4.2 continued from the previous page.

Setting	Value
Teacher weighting strategy	routing (GRACE path)
Distillation loss	Trie-Wasserstein
Trie-Wasserstein rho	0.7
Trie-Wasserstein top- k	64
Training dataset	DocVQA
Validation split	validation
Hardware and runtime	Single-GPU runs on Modal (H200) or Vast.ai
Typical run cost	About 7 hours on H200, about 14.41 USD at 2.059 US-D/hour on Vast.ai
Distributed training	DeepSpeed ZeRO-2
Optimizer	adamw_torch
Adam betas	(0.9, 0.98)
Adam epsilon	1×10^{-7}
Weight decay	0.0
Numerical precision	bf16 enabled, half precision (fp16) disabled, TF32 enabled
FlashAttention-2	Enabled
Training data format	LLaVA-style multimodal instruction JSON
Max sequence length	16384
Training epochs	2
Per-device batch size	80
Gradient accumulation steps	1
Effective batch size	80
Base learning rate	1×10^{-5}
Vision learning rate	2×10^{-6}
Connector learning rate	1×10^{-5}
Learning-rate schedule	Cosine decay with warmup ratio 0.03
Gradient checkpointing	Enabled
Frozen modules	None
Dataloader workers	4
Logging and checkpointing	Log every 5 steps and save every 150 steps
Student/teacher temperature	$\tau_s = \tau_t = 1.0$
KD weight	$\alpha_{\text{KD}} = 0.5$
Top- k routing	$k_{\text{route}} = 2$, with top-scoring teacher masking and selected softmax-weight renormalization
Gate entropy weight	$\lambda_{\text{ent}} = 10^{-3}$
Router z -loss weight	$\lambda_z = 10^{-3}$
GRACE warmup ratio	$\rho_g = 0.05$

Continued on the next page.

Table 4.2 continued from the previous page.

Setting	Value
GRACE softmax beta	$\beta_{\text{score}} = 4.0$
GRACE blend exponent	$\lambda_{\text{blend}} = 0.9$
Score-spread fallback threshold	$\epsilon_{\text{fb}} = 0.01$
Alignment threshold	$\delta = 0.15$
Seed	42
Image resolution / visual token budget	Processor defaults with no manual override

GRACE score smoothing. GRACE applies an exponential moving average to teacher-gradient agreement scores before final teacher weighting. Section 5.3 evaluates the effect of removing this smoothing with the $\beta_{\text{EMA}} = 0$ ablation.

4.4 Evaluation Metrics

We report benchmark-specific official metrics for each dataset.

TextVQA. We report VQA Accuracy. For a predicted answer $\hat{a}$, the per-question score is

$$s_{\text{VQA}} = \min \left(\frac{\# \text{annotators that provided } \hat{a}}{3}, 1 \right),$$

and the final score is the mean over all questions [7].

DocVQA. We report ANLS, the Average Normalized Levenshtein Similarity. The final ANLS is the average score over all samples [8]:

$$\text{NLS}(a, \hat{a}) = 1 - \frac{d_{\text{Lev}}(a, \hat{a})}{\max(|a|, |\hat{a}|)}, \quad s_{\text{ANLS}} = \begin{cases} \text{NLS}(a, \hat{a}), & \text{NLS} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

ChartQA. We report Relaxed Accuracy. A prediction is correct if it matches exactly for text answers. For numeric answers, the score handles the zero-target case explicitly:

$$s_{\text{ChartQA}} = \begin{cases} \mathbf{1}[\hat{y} = y], & y = 0, \\ \mathbf{1}\left[\frac{|\hat{y}-y|}{|y|} \leq 0.05\right], & y \neq 0. \end{cases}$$

The final ChartQA score is computed as the proportion of predictions that satisfy either the exact text-match criterion or the relaxed numeric-match criterion [9].

Table 4.3: Baseline groups used in the evaluation chapter. The grouping separates student-only references, same-family KD, tokenizer-mismatched single-teacher transfer and heterogeneous teacher methods.

Group	Methods	Teacher setup	Purpose
Non-distillation references	Original, full finetuning	No teacher	Student-only reference points
Same-family baselines	Forward KL, Reverse KL, Jensen-Shannon	SmolVLM2-2.2B-Instruct	Standard single-teacher objectives within the SmolVLM family
Different-family single-teacher baselines	ULD	Qwen2-VL-2B-Instruct	Cross-family transfer without routed teacher selection
Heterogeneous teacher baselines	Uniform averaging, SHM-CKA, REINFORCE, GRACE (Ours)	Mixed teacher set from multiple VLM families	Static averaging vs. hidden-state matching vs. adaptive teacher assignment

4.5 Baselines and Compared Methods

For consistency with Chapter 5, we organize the compared methods into four groups: non-distillation references, same-family distillation baselines, different-family single-teacher baselines and heterogeneous teacher baselines.

Before introducing the method-specific baselines, Table 4.3 summarizes the role of each baseline family and separates student-only references, same-family KD, tokenizer-mismatched single-teacher transfer and heterogeneous teacher methods.

The non-distillation references consist of the original student and the full-finetuning baseline, which provide student-only reference points. The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon distillation, isolating standard teacher-student transfer when the tokenizer and output space are matched [16, 21, 48, 59, 60]. The different-family single-teacher setting uses ULD with a teacher outside the SmolVLM family, testing tokenizer-mismatched transfer without teacher routing [25]. The heterogeneous teacher setting compares uniform averaging, SHM-CKA, REINFORCE and GRACE over a mixed teacher pool, separating static averaging, hidden-state matching and adaptive teacher assignment [26, 27, 34, 40].

For the trainable competitor methods, we keep the supervised term $\mathcal{L}_{CE}$. The logits-only baselines differ only in the KD signal, but SHM-CKA also adds a hidden-state alignment term and REINFORCE also adds a REINFORCE-style policy term [33, 39, 61]. In the descriptions below, γ denotes a generic KD mixing coefficient. α remains reserved for the repository training script in Table 4.2.

Original. The original setting directly evaluates the released student without additional optimization, thereby measuring the native capability of the student model before task adaptation or distillation.

Full finetuning. Full finetuning optimizes only $\mathcal{L}_{CE}$ and uses no teacher, which isolates the effect of supervised task adaptation from any teacher-guided distillation signal.

Forward KL. Forward KL uses the standard same-family KD objective

$$\mathcal{L}_{KD} = \text{KL}(p_t \| p_s).$$

This tests soft-target transfer in the matched-tokenizer setting where teacher and student vocabularies are aligned [21].

Reverse KL. Reverse KL uses $\mathcal{L}_{KD} = \text{KL}(p_s \| p_t)$ [60, 62] as a student-centered divergence baseline, which tests whether matching the teacher from the student’s current distribution is suitable for VLM distillation. Because the reverse-KL gradient is weighted by p_s , this objective tends to emphasize regions where the student already assigns probability.

Jensen-Shannon. Jensen-Shannon uses the symmetric divergence

$$\frac{1}{2}\text{KL}(p_s \| m) + \frac{1}{2}\text{KL}(p_t \| m), \quad m = \frac{1}{2}(p_s + p_t).$$

It tests whether a bounded symmetric objective is more stable than KL [48, 60].

ULD. ULD uses $\mathcal{L}_{KD} = \|\pi(p_s) - \pi(p_t)\|_1$ [25]. This is the main tokenizer-mismatched logits baseline because it compares ranked probability mass, not token IDs.

SHM-CKA. SHM-CKA keeps the heterogeneous teacher set used by GRACE, but replaces routed weighting with a per-teacher ULD logits loss plus a hidden-state CKA penalty [25, 27, 39]. It tests whether direct representation alignment is sufficient without routed GRACE weighting. This baseline separates representation matching from output-level teacher selection.

Uniform multi-teacher. Uniform multi-teacher distillation assigns fixed equal weights to all teachers in the heterogeneous pool, $\mathcal{L}_{KD} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{KD}^{(i)}$ [33, 34], thereby testing whether teacher diversity alone is sufficient without adaptive routing or gradient-aware weighting.

REINFORCE. REINFORCE replaces deterministic routing with a stochastic policy that samples teachers and updates the selector with REINFORCE [33, 61], providing an adaptive routing baseline that does not use the gradient-agreement refinement introduced by GRACE.

For SHM-CKA, the baseline implementation first loads an offline CKA matrix and, for each student layer s , selects the top- k teacher layers $\mathcal{T}_s$. Their CKA scores are normalized into soft weights $\alpha_{s,t}$. The per-teacher layer loss is

$$\mathcal{L}_{\text{layer}}^{(i)} = \frac{1}{|S|} \sum_{s \in S} \sum_{t \in \mathcal{T}_s} \alpha_{s,t} (1 - \sqrt{\text{CKA}(\mathbf{H}_s, \mathbf{H}_t)}). \quad (4.1)$$

Each teacher also contributes its own ULD logits loss $\mathcal{L}_{\text{ULD}}^{(i)}$. In the original single-teacher implementation, the default fixed-weight objective gives equal weight to CE, logits KD and layer matching:

$$\mathcal{L}_{\text{fixed}} = \frac{1}{3} (\mathcal{L}_{CE} + \mathcal{L}_{\text{ULD}} + \mathcal{L}_{\text{layer}}). \quad (4.2)$$

That implementation also supports an alternative adaptive weighting scheme, but the fixed equal-mean form above is the default setting. The original soft top- k matching implementation was designed for single-teacher runs. For the heterogeneous baseline used here, we preserve the hidden-state matching idea for each teacher and average the logits and layer-matching terms across teachers:

$$\bar{\mathcal{L}}_{\text{ULD}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{ULD}}^{(i)}, \quad \bar{\mathcal{L}}_{\text{layer}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{layer}}^{(i)}, \quad (4.3)$$

then applies an equal-mean combination

$$\mathcal{L}_{\text{SHM-CKA}} = \frac{1}{3} (\mathcal{L}_{CE} + \bar{\mathcal{L}}_{\text{ULD}} + \bar{\mathcal{L}}_{\text{layer}}). \quad (4.4)$$

For REINFORCE, the policy uses the pooled student hidden state as a sample descriptor, then concatenates per-teacher summary features:

$$\mathbf{z} = \text{Concat}(\text{Pool}(\mathbf{H}_s), \mathbf{f}_1, \dots, \mathbf{f}_M), \quad \mathbf{f}_i = [\mathcal{L}_{KD}^{(i)}, c_i, e_i], \quad (4.5)$$

where c_i is the teacher’s mean token confidence and e_i is its normalized token entropy over supervised positions. A linear policy then predicts independent Bernoulli probabilities

$$\boldsymbol{\pi} = \sigma(W \text{LN}(\mathbf{z}) + b). \quad (4.6)$$

During the warmup stage, all teachers stay active. After warmup, the baseline samples $m_i \sim \text{Bernoulli}(\pi_i)$, falls back to the highest-probability teacher if $\sum_i m_i = 0$ and normalizes the selected mask into weights

$$w_i = \frac{m_i}{\sum_{j=1}^M m_j}. \quad (4.7)$$

The multi-teacher KD term is the weighted average

$$\mathcal{L}_{KD} = \sum_{i=1}^M w_i \mathcal{L}_{KD}^{(i)}. \quad (4.8)$$

In the REINFORCE training configuration, $\mathcal{L}_{KD}^{(i)}$ is the per-teacher trie-Wasserstein loss. The selector is trained with a REINFORCE-style term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (4.9)$$

where b is an EMA reward baseline. The REINFORCE training configuration uses $r = -\mathcal{L}_{CE} - \frac{1}{\sum_i m_i} \sum_i m_i \mathcal{L}_{KD}^{(i)}$, so the total reinforced-selection objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (4.10)$$

Unless noted otherwise, all baselines reuse the data conversion pipeline, hardware, optimizer family and checkpointing protocol described earlier in this chapter. Methods that do not use routing simply omit the GRACE-specific terms and retain the corresponding single-teacher or uniform multi-teacher objective from the cited source.

4.6 Implementation Details

This section records the engineering details needed to reproduce the main training configuration. These details are useful for systems reproducibility, but they are separate from the fair-comparison protocol summarized earlier.

4.6.1 Offline Teacher-Logit Preparation

Teacher logits are generated *offline* before distillation. An offline caching script runs each frozen teacher on every training example, keeps only the supervised answer-token positions and writes one `.pt` file per sample under a teacher-specific directory. Each record stores the sample index, the selected teacher logits and the corresponding supervised labels. This keeps the online training loop simple: the student only reads precomputed teacher targets instead of running the teachers again.

After generation, the teacher-logit directories are uploaded to a **Cloudflare R2** bucket using its S3-compatible object-storage interface [63, 64]. In the active setup, the bucket already exceeds **1 TB**, so copying the full cache to an external GPU instance is impractical. This is especially relevant on services such as Vast.ai, where local NVMe space is usually much smaller than the full teacher logit store.

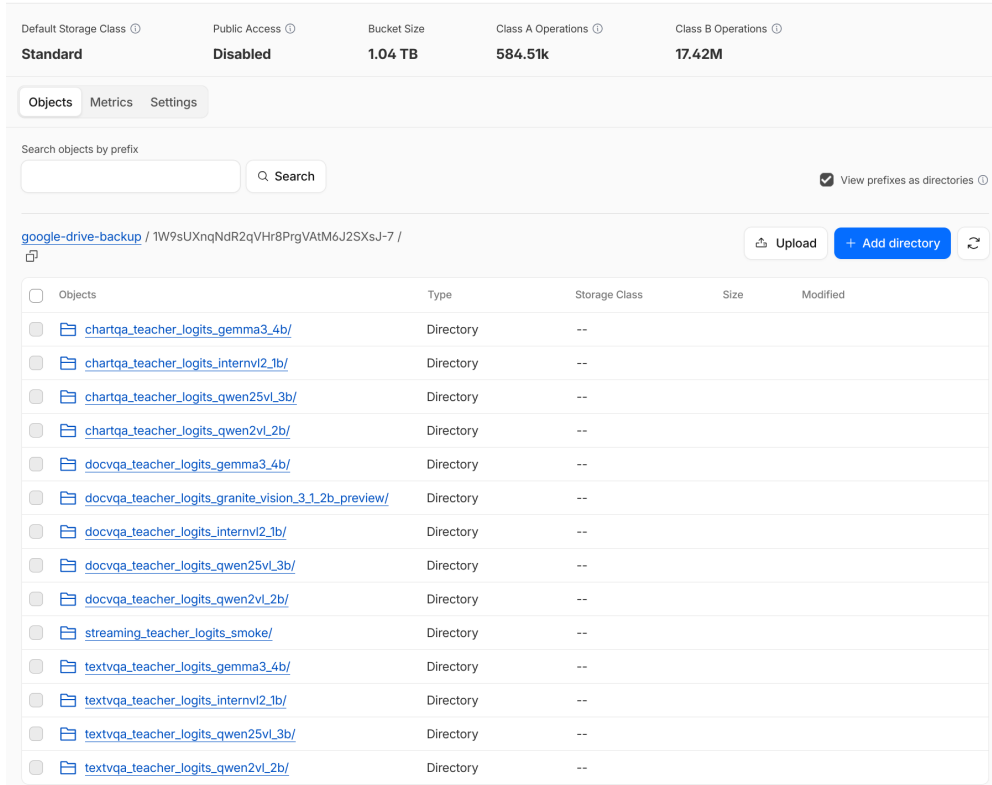


Figure 4.2: Cloudflare R2 bucket used for offline teacher-logit storage.

The remote bucket layout used for the offline teacher caches appears in Figure 4.2.

4.6.2 Remote Logit Streaming During Training

The main training path does *not* download the full cache before training. When the launcher receives an R2 root in `distill_multi_teachers.sh`, the trainer switches to remote lazy loading. During training, `StreamingTeacherLogitsCache` resolves the teacher-specific remote object path

`remote_root/<dataset>_teacher_logits_<alias>/<teacher-slug>/<index>.pt`,

then fetches files lazily from R2 through `s3fs` and `WholeFileCacheFileSystem` [65, 66]. The repository also includes a repacking utility for MDS, but the active training path reads the raw per-sample `.pt` cache directly.

To keep disk usage bounded, the loader uses a **100 GB** local cache. The loader caches each file only after first access, updates its access time after opening and removes the oldest files first once the cache exceeds the limit. In effect, the local cache behaves as a small least-recently-used style buffer over the much larger R2 store. A per-teacher file lock also guards cache metadata updates, so concurrent dataloader workers do not corrupt the cache state.

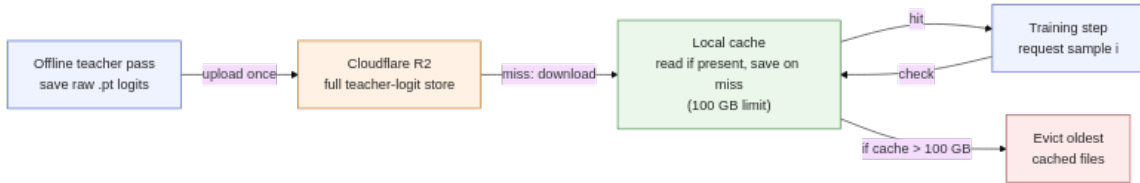


Figure 4.3: Training with remote teacher logits.

The remote-cache training path appears in Figure 4.3.

Table 4.4: Engineering settings for the active four-teacher DocVQA logit-streaming pipeline. These values describe storage, caching and launch-path behavior, not the fair-comparison protocol itself.

Setting	Value
Teacher-logit storage	Cloudflare R2 object storage
Teacher-logit format	Raw per-sample <code>.pt</code> files
Remote path template	<code>remote_root/<dataset>_teacher_logits_<alias>/<teacher-slug>/<index>.pt</code>
Remote filesystem stack	<code>s3fs</code> + <code>WholeFileCacheFileSystem</code>
Optional repacking utility	MDS repacking is available but not used in the active path

Continued on the next page.

Table 4.4 continued from the previous page.

Setting	Value
Local cache directory	<code>/tmp/teacher-logits-streaming</code> for R2 reads <code>/workspace/cache</code> for local-cache reads
Local cache limit	100 GB
Cache eviction rule	Remove the oldest cached files first
Cache synchronization	Per-teacher file lock on cache metadata updates

Chapter 5

Experiment Results

5.1 Main Comparison Results

This section reports the main empirical comparison used to answer RQ1. The results are organized from simpler reference points to the full heterogeneous multi-teacher setting: original students, supervised fine-tuning, same-family logits KD, cross-family ULD and multi-teacher distillation methods.

The section first establishes the student and teacher reference ranges, then compares all methods on TextVQA, DocVQA and ChartQA. This ordering separates two questions: whether distillation improves over the student-only reference point and whether adaptive multi-teacher weighting improves over fixed or single-teacher transfer.

5.1.1 Baseline Results

Before applying adaptation or distillation, we report the original performance of the two student models, **SmolVLM-500M** and **SmolVLM-256M**, using **VLM-Eval-Kit** [53] under the evaluation pipeline described in Chapter 4. Table 5.1 gives the student-only starting point for all later fine-tuning and distillation comparisons. Table 5.2 then reports the teacher-side reference range. The gap between these student models and the stronger teachers motivates the distillation experiments that follow.

5.1.2 Overall Comparison

Tables 5.3, 5.4 and 5.5 summarize the main comparison results across original models, direct fine-tuning, standard logits distillation baselines and GRACE.

Unless stated otherwise, the main comparison tables report the strongest completed run for each method and benchmark. Uniform, SHM-CKA, REINFORCE and GRACE use a four-teacher mixed set spanning the Qwen, InternVL and Gemma families. Chapter 4 documents the active four-teacher DocVQA launch script for system-level reproducibility.

Table 5.1: Original benchmark results of the two SmolVLM students on TextVQA, DocVQA and ChartQA. These scores are the student-only reference point for all later fine-tuning and distillation comparisons.

Model	TextVQA	DocVQA	ChartQA
SmolVLM-500M	56.75	59.52	62.80
SmolVLM-256M	47.17	48.64	45.04

Table 5.2: Original benchmark results of the teacher and large reference models used here. The table establishes the teacher-side reference range used later in the gap-to-teacher analysis.

Teacher Model	TextVQA	DocVQA	ChartQA
Qwen2.5-VL-3B-Instruct	79.30	93.90	94.24
InternVL3-1B	74.10	81.90	75.30
Gemma-3-4B-it	57.80	75.80	63.60
Qwen2-VL-2B-Instruct	79.70	90.10	89.28

We report the baselines in four groups. The non-distillation references are the original student and full-finetuning runs. The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon divergence [21, 22, 27]. The different-family single-teacher setting uses ULD with a teacher outside the SmolVLM family [25]. The multi-teacher baselines compare four-teacher Uniform averaging with adaptive methods such as SHM-CKA, REINFORCE and GRACE [33, 34, 40, 61].

Because each multi-teacher run requires large offline logit caches and long single-GPU training, we report one fully trained run per method. A typical H200 run takes about 7 hours (about 14.41 USD at 2.059 USD/hour on Vast.ai before storage and offline-logit generation cost). We therefore treat small differences cautiously and focus on consistent multi-point gaps across datasets and student sizes.

On DocVQA, GRACE rises from 69.59 to 73.11 on SmolVLM-500M and from 55.70 to 60.48 on SmolVLM-256M relative to the Uniform baseline.

ChartQA shows the largest margins over Uniform: 75.60 to 80.20 for SmolVLM-500M and 63.68 to 72.32 for SmolVLM-256M.

Figure 5.4 provides one illustrative optimization trace.

Table 5.3: TextVQA results for the original students, fine-tuning, single-teacher baselines and heterogeneous teacher methods. Uniform, SHM-CKA, REINFORCE and GRACE use the same four-teacher pool.

Category	Method	Teacher	Student	TextVQA	
Non-distillation references					
Original	None	None	SmolVLM-500M	56.75	
			SmolVLM-256M	47.17	
Reference	Finetuning	None	SmolVLM-500M	56.96	
			SmolVLM-256M	49.83	
Same-family distillation baselines					
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	59.00	
			SmolVLM-256M	50.88	
	Reverse KL		SmolVLM-500M	44.73	
			SmolVLM-256M	29.75	
	Jensen-Shannon		SmolVLM-500M	42.33	
			SmolVLM-256M	33.84	
Different-family single-teacher baselines					
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	61.49	
			SmolVLM-256M	53.95	
Multi-teacher baselines					
	Uniform		SmolVLM-500M	63.88	
			SmolVLM-256M	54.57	
	SHM-CKA		Qwen2-VL-2B-Instruct	SmolVLM-500M	60.47
			Qwen2.5-VL-3B-Instruct	SmolVLM-256M	55.93
	REINFORCE		InternVL3-1B	SmolVLM-500M	63.26
			Gemma-3-4B-it	SmolVLM-256M	56.31
	GRACE (Ours)		SmolVLM-500M	69.38	
			SmolVLM-256M	60.73	

Table 5.4: DocVQA results for the original students, fine-tuning, single-teacher baselines and heterogeneous teacher methods. Uniform, SHM-CKA, REINFORCE and GRACE use the same four-teacher pool.

Category	Method	Teacher	Student	DocVQA	
Non-distillation references					
Original	None	None	SmolVLM-500M	59.52	
			SmolVLM-256M	48.64	
Reference	Finetuning	None	SmolVLM-500M	61.96	
			SmolVLM-256M	42.03	
Same-family distillation baselines					
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	63.55	
			SmolVLM-256M	49.46	
	Reverse KL		SmolVLM-500M	29.28	
			SmolVLM-256M	23.28	
	Jensen-Shannon		SmolVLM-500M	36.43	
			SmolVLM-256M	33.12	
Different-family single-teacher baselines					
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	65.28	
			SmolVLM-256M	52.24	
Multi-teacher baselines					
	Uniform		SmolVLM-500M	69.59	
			SmolVLM-256M	55.70	
	SHM-CKA		Qwen2-VL-2B-Instruct	SmolVLM-500M	63.32
			Qwen2.5-VL-3B-Instruct	SmolVLM-256M	57.24
	REINFORCE		InternVL3-1B	SmolVLM-500M	64.00
			Gemma-3-4B-it	SmolVLM-256M	58.53
	GRACE (Ours)		SmolVLM-500M	73.11	
			SmolVLM-256M	60.48	

Table 5.5: ChartQA results for the original students, fine-tuning, single-teacher baselines and heterogeneous teacher methods. Uniform, SHM-CKA, REINFORCE and GRACE use the same four-teacher pool.

Category	Method	Teacher	Student	ChartQA
Non-distillation references				
Original	None	None	SmolVLM-500M	62.80
			SmolVLM-256M	45.04
Reference	Finetuning	None	SmolVLM-500M	63.00
			SmolVLM-256M	49.68
Same-family distillation baselines				
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	64.52
			SmolVLM-256M	56.80
	Reverse KL		SmolVLM-500M	30.16
			SmolVLM-256M	17.20
	Jensen-Shannon		SmolVLM-500M	43.75
			SmolVLM-256M	50.93
Different-family single-teacher baselines				
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	66.92
			SmolVLM-256M	60.96
Multi-teacher baselines				
	Uniform	Qwen2-VL-2B-Instruct Qwen2.5-VL-3B-Instruct InternVL3-1B Gemma-3-4B-it	SmolVLM-500M	75.60
			SmolVLM-256M	63.68
	SHM-CKA		SmolVLM-500M	76.88
			SmolVLM-256M	62.40
	REINFORCE		SmolVLM-500M	70.27
			SmolVLM-256M	68.74
	GRACE (Ours)		SmolVLM-500M	80.20
			SmolVLM-256M	72.32

Several trends are consistent across the three benchmarks. Same-family logits baselines remain weak: forward KL gives only small gains, reverse KL is consistently poor and Jensen-Shannon is unstable. Cross-family ULD is already stronger than the same-family logits losses on all three tasks, showing that tokenizer-mismatched distillation is more useful here than vocabulary-matched KL. Once multiple teachers are added, Uniform averaging gives another clear gain, which means teacher diversity itself is valuable. The additional improvement from single-teacher ULD to GRACE comes from the full recipe: trie-Wasserstein KD, sample-dependent sparse routing and gradient-agreement refinement after routing. GRACE also improves on the uniform multi-teacher baseline for both student sizes. The gain is moderate on TextVQA and DocVQA, but larger on ChartQA, indicating that adaptive weighting helps most when the task mixes visual reading, numerical reasoning and teacher specialization.

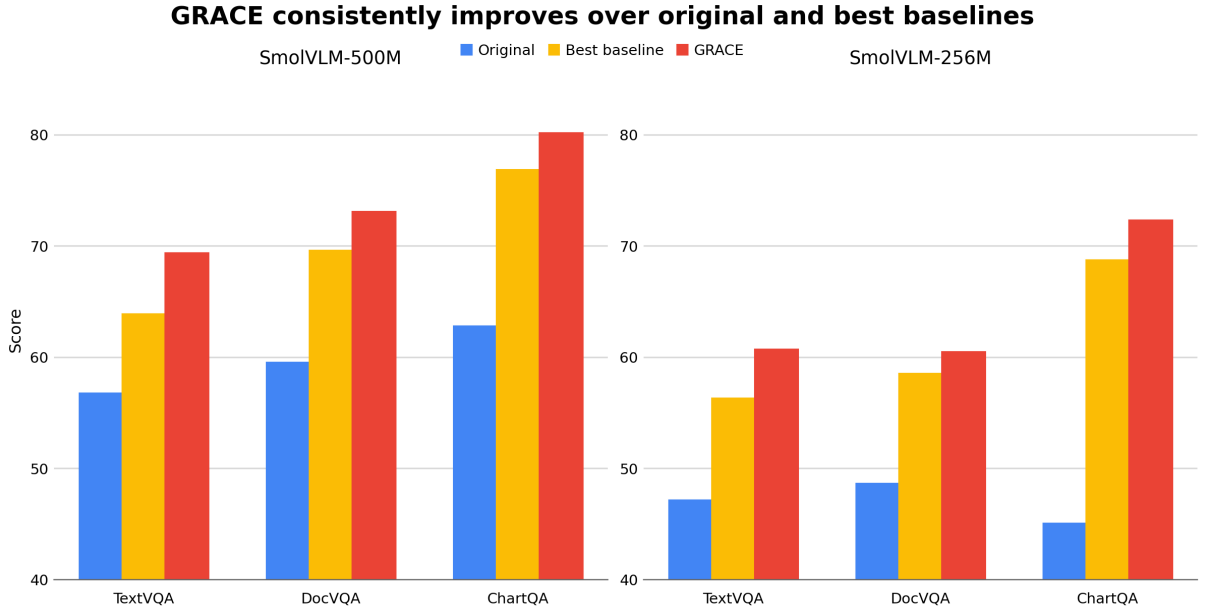


Figure 5.1: Grouped comparison of the original student, the strongest non-GRACE baseline and GRACE for both student sizes.

Figure 5.1 summarizes the main ranking by showing that GRACE remains above the strongest non-GRACE baseline across all three benchmarks and both student sizes.

For the qualitative analysis below, we use DEX-AR [67]. For token t , layer l and head h , DEX-AR forms a gradient-based attribution map

$$\mathbf{G}_t^{(l,h)} = \nabla_{\mathbf{A}^{(l,h)}} y_t \odot \mathbf{A}^{(l,h)},$$

then keeps visually grounded heads with

$$w^{(l,h)} = \max(S_{\text{img}}^{(l,h)} - S_{\text{text}}^{(l,h)}, 0),$$

and aggregates

$$\mathbf{H}_t = \sum_{l,h} w^{(l,h)} \mathbf{G}_t^{(l,h)}, \quad \mathbf{H}_{\text{sent}} = \sum_t \delta_t \mathbf{H}_t.$$

Here $\delta_t = 1/|\Omega_{\text{ans}}|$ for answer tokens $t \in \Omega_{\text{ans}}$ and $\delta_t = 0$ otherwise, so the sentence-level map is the uniform average of token-level maps over the generated answer span. We prefer DEX-AR over raw attention maps or generic saliency because the model is autoregressive and the target is answer-token grounding, not generic image saliency. Figure 5.2 summarizes this attribution pipeline before the qualitative TextVQA example.

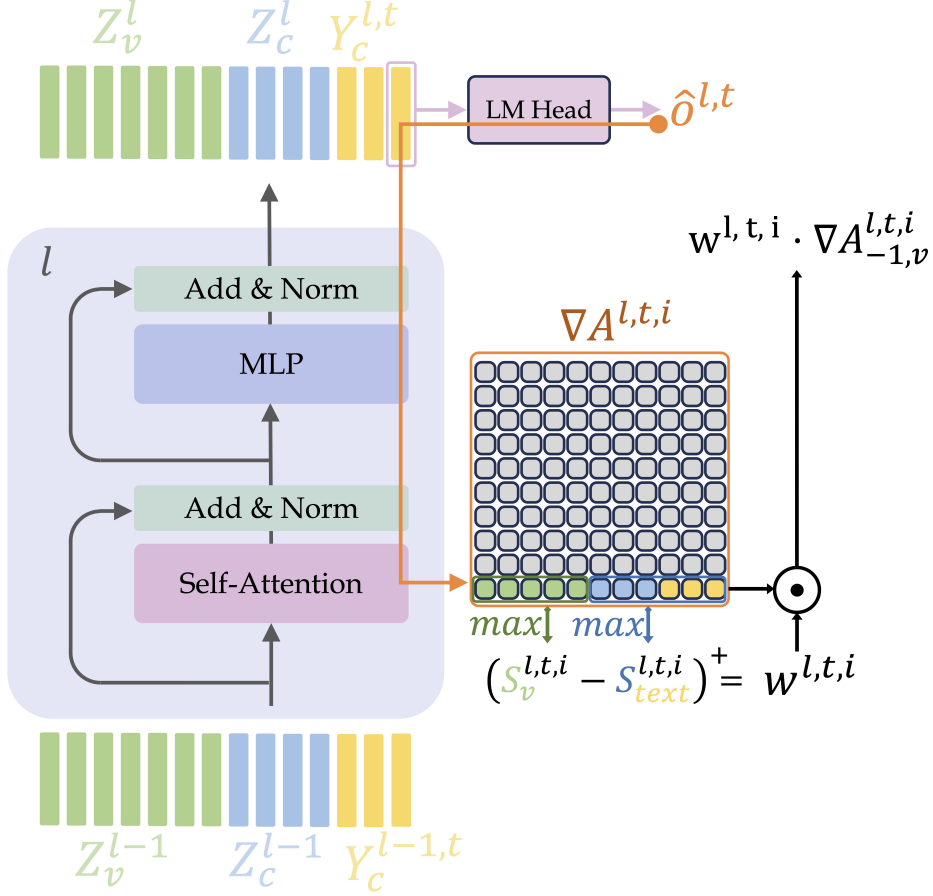


Figure 5.2: DEX-AR overview. The method computes gradient-based attribution over autoregressive attention maps, filters visually grounded heads and aggregates token-level maps into a sentence-level explanation.

Figure 5.3 gives a representative TextVQA example. Sample 3 asks “*who is this copyrighted by?*” The relevant visual evidence is the copyright text on the lower-left border. The original SmolVLM-500M instead assigns high attention to the *Virgin* logo, which is not the region needed to answer the question. Compared with the original student, the GRACE model reduces this attention error by shifting more visual evidence toward the copyright-text region needed to answer the question.



(a) Input image



(b) Original SmolVLM-500M



(c) GRACE SmolVLM-500M

Figure 5.3: Representative TextVQA example for sample 3 (“*who is this copyrighted by?*”). The original SmolVLM-500M places highest mass on the salient *Virgin* logo, while the GRACE model shifts attention toward the copyright-text region.

Answer to RQ1

GRACE improves over direct finetuning, same-family logits KD, ULD and the compared heterogeneous multi-teacher baselines. The ranking is clear: same-family logits baselines are weakest, cross-family ULD is stronger, multi-teacher baselines improve further and GRACE is best overall. Within the multi-teacher group, static Uniform averaging already helps, but GRACE stays ahead of SHM-CKA, REINFORCE and Uniform itself. This shows that teacher diversity already helps, but the largest gains come from combining the trie-Wasserstein loss with gradient-based alignment. On SmolVLM-500M, GRACE closes 55.03%, 39.53% and 55.34% of the gap between the strongest teacher in the four-teacher comparison pool and the student on TextVQA, DocVQA and ChartQA, respectively.

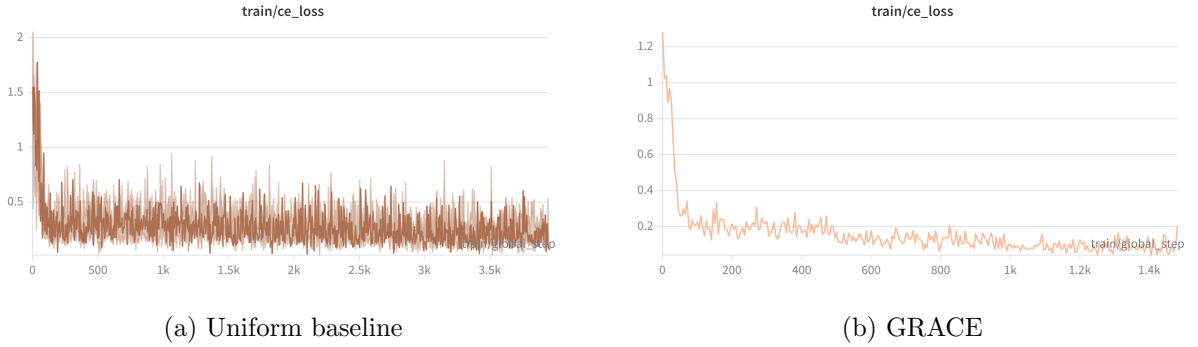


Figure 5.4: Training curves from a Uniform run and one representative GRACE run in the DocVQA experiment.

Looking at Figure 5.4, the GRACE run has a much smoother and more stable training curve than the Uniform baseline in the DocVQA experiment. The Uniform curve shows sharper oscillations, while the GRACE curve decreases more steadily after the early training phase.



(a) Without GRACE refinement



(b) With GRACE

Figure 5.5: Representative routed-teacher load plots from one collapse case and one stable GRACE case.

Figure 5.5 contrasts collapsed and stable routing behavior. Without the full GRACE recipe, the routed load concentrates on one teacher, reducing teacher diversity. With GRACE, the load stays distributed across the teacher pool, so the student continues to receive heterogeneous supervision.

Table 5.6 states the result numerically for the SmolVLM-500M student. Relative gain is computed as $(\text{GRACE} - \text{baseline})/\text{baseline}$.

Table 5.6: Absolute and relative gains of GRACE on SmolVLM-500M against the original model, direct finetuning, uniform weighting and the strongest non-GRACE baseline. Relative gain is computed as $(\text{GRACE} - \text{baseline})/\text{baseline}$.

Benchmark	vs Original	vs Finetuning	vs Uniform	vs Best non-GRACE
TextVQA	+12.63 (+22.26%)	+12.42 (+21.80%)	+5.50 (+8.61%)	+5.38 (+8.41%)
DocVQA	+13.59 (+22.83%)	+11.15 (+18.00%)	+3.52 (+5.06%)	+3.52 (+5.06%)
ChartQA	+17.40 (+27.71%)	+17.20 (+27.30%)	+4.60 (+6.08%)	+3.32 (+4.32%)

Table 5.7: Gap-to-teacher analysis for SmolVLM-500M under GRACE. The teacher reference is the strongest model inside the four-teacher GRACE comparison pool for each benchmark.

Benchmark	Best teacher	Original gap	GRACE gap	Gap closed
TextVQA	79.70	22.95	10.32	55.03%
DocVQA	93.90	34.38	20.79	39.53%
ChartQA	94.24	31.44	14.04	55.34%

The gains over the original SmolVLM-500M student are large: +22.26% on TextVQA, +22.83% on DocVQA and +27.71% on ChartQA. The gains over direct finetuning are almost as large, at +21.80%, +18.00% and +27.30%, respectively. The gain over static uniform averaging is smaller, but it remains consistent across all three tasks, isolating the value of adaptive teacher weighting from the value of simply using more than one teacher. The pattern also holds for SmolVLM-256M: GRACE rises from 47.17 to 60.73 on TextVQA (+28.75%), from 48.64 to 60.48 on DocVQA (+24.34%) and from 45.04 to 72.32 on ChartQA (+60.57%). Relative to the uniform baseline, the SmolVLM-256M gains +11.29% on TextVQA, +8.58% on DocVQA and +13.57% on ChartQA.

The trend is similar if the reference point is the strongest teacher in the GRACE comparison pool reported in Table 5.2. Let $g_{\text{orig}} = y_T - y_{\text{orig}}$ and $g_{\text{grace}} = y_T - y_{\text{grace}}$. The fraction of the gap closed is $1 - g_{\text{grace}}/g_{\text{orig}}$.

The gain is not merely an artifact of comparison with weak baselines. On two benchmarks, GRACE removes about half of the original teacher-student gap and even on DocVQA, where the teacher ceiling is highest, it still closes almost 40% of the gap.

The distilled student can outperform individual teachers. On TextVQA, GRACE reaches 69.38 and exceeds Gemma-3-4B-it at 57.80 by +11.58 points (+20.03%). On ChartQA, it exceeds InternVL3-1B at 75.30 by +4.90 points (+6.51%) and Gemma-3-4B-it at 63.60 by +16.60 points (+26.10%). This suggests that routed multi-teacher supervision can combine useful signals that are not available from any single teacher alone.

One negative result is also consistent: reverse KL is much weaker than forward KL. This can be explained by the different student-logit gradients. Let $p_s = \text{softmax}(z_s)$ be the student distribution and let p_t be the fixed teacher distribution. For forward KL,

$$\mathcal{L}_{FKL} = \text{KL}(p_t \| p_s) = \sum_v p_t(v) \log \frac{p_t(v)}{p_s(v)}, \quad \frac{\partial \mathcal{L}_{FKL}}{\partial z_s(v)} = p_s(v) - p_t(v).$$

Thus, every teacher-supported token directly pulls the student distribution toward the teacher’s soft target [21]. Reverse KL gives

$$\mathcal{L}_{RKL} = \text{KL}(p_s \| p_t) = \sum_v p_s(v) \log \frac{p_s(v)}{p_t(v)}.$$

The student-logit gradient is

$$\frac{\partial \mathcal{L}_{RKL}}{\partial z_s(v)} = p_s(v) \left(\log \frac{p_s(v)}{p_t(v)} - \mathcal{L}_{RKL} \right).$$

The key difference is the factor $p_s(v)$. If $p_t(v) > 0$ but $p_s(v) \rightarrow 0$, then

$$\frac{\partial \mathcal{L}_{FKL}}{\partial z_s(v)} \rightarrow -p_t(v), \quad \frac{\partial \mathcal{L}_{RKL}}{\partial z_s(v)} \rightarrow 0.$$

Teacher-supported tokens that the student currently assigns low probability therefore receive weaker correction under reverse KL. In finite training, this student-weighted gradient can make reverse KL less useful for answer-token distillation, especially when teacher distributions contain useful secondary mass over OCR variants, numeric formats or plausible answer alternatives. We interpret this as an optimization-bias explanation, not as a universal rule, because recent LLM distillation work shows that the simple mode-seeking interpretation of reverse KL does not always fully describe KD behavior [68].

SHM-CKA is also not sufficient by itself because representation similarity is not the same as task-useful output supervision. CKA measures hidden-state geometry [39], but hidden-state similarity does not imply output equivalence:

$$\text{CKA}(H_s, H_t) \approx 1 \not\Rightarrow p_s(\cdot | x) \approx p_t(\cdot | x).$$

A simple linear-head example shows why. Suppose the teacher representation is an orthogonal rotation of the student representation, $H_t = H_s R$ with $R^\top R = I$. Linear CKA gives

$$\text{CKA}(H_s, H_t) = \frac{\|H_t^\top H_s\|_F^2}{\|H_t^\top H_t\|_F \|H_s^\top H_s\|_F} = 1.$$

However, the answer logits can still differ:

$$z_s = H_s W_s, \quad z_t = H_t W_t = H_s R W_t.$$

Unless $W_s = RW_t$, the two models can have identical CKA but different logits and different answer-token distributions. This distinction matches the difference between representational similarity and functional similarity discussed in prior work [69]. In heterogeneous VLM distillation, the final answer distribution also depends on the language head, tokenizer, output vocabulary and task-specific teacher behavior. Although SHM-CKA is competitive on ChartQA, it remains weaker than GRACE on TextVQA and DocVQA, where output-level tokenizer-agnostic matching and sample-dependent teacher weighting are more important.

A clear limitation remains on the vision side because the current method distills answer-token output distributions but does not explicitly align visual tokens, cross-attention maps or region-level evidence. The method pools student hidden states for routing and scores teachers from CE/KD agreement at the output side, which means the vision encoder is treated mostly through its effect on the final answer distribution, not as a first-class distillation target.

5.2 Representation Similarity Analysis

We use Centered Kernel Alignment (CKA) as an analysis tool to examine whether teacher and student layers exhibit clear correspondence. For centered hidden-state matrices $\tilde{\mathbf{H}}_s$ and $\tilde{\mathbf{H}}_t$, linear CKA is

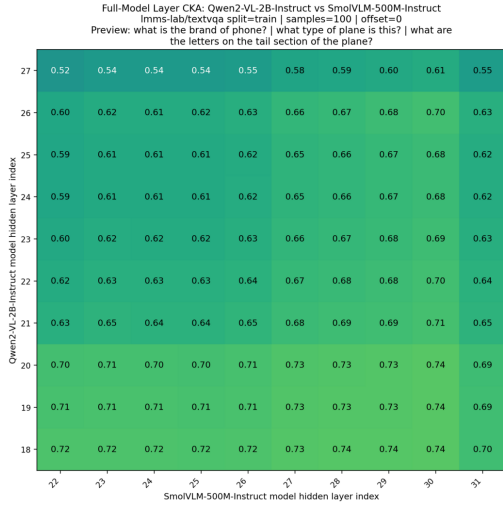
$$\text{CKA}(\tilde{\mathbf{H}}_s, \tilde{\mathbf{H}}_t) = \frac{\left\| \tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_s \right\|_F^2}{\left\| \tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_t \right\|_F \left\| \tilde{\mathbf{H}}_s^\top \tilde{\mathbf{H}}_s \right\|_F}. \quad (5.1)$$

Values closer to 1 indicate stronger representation similarity [39]. CKA is therefore used here as a representation-analysis tool, not as evidence that two models implement the same answer function. A high CKA score indicates similar hidden geometry on the evaluated samples, but it does not guarantee similar answer-token probabilities after the language head.

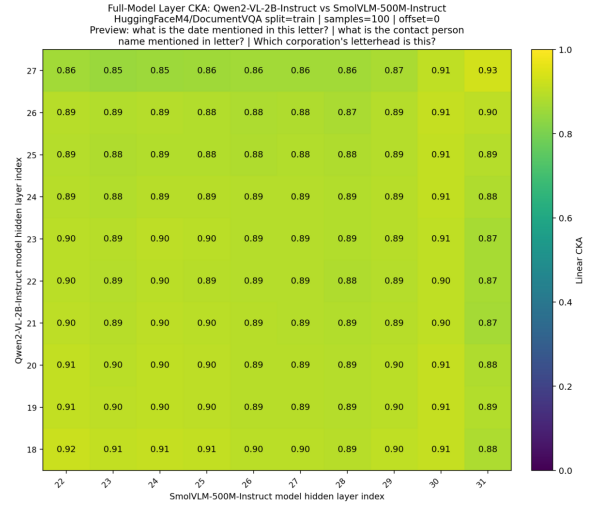
The main question is whether strong internal similarity is enough to justify layer-wise distillation. Our results suggest that it is not, as shown in the figures and tables below.

The final-layer maps show that the teacher and student are not severely misaligned in representation space.

Teacher and student can remain relatively close in both the language and vision stacks, as Figures 5.6 and 5.7 indicate.

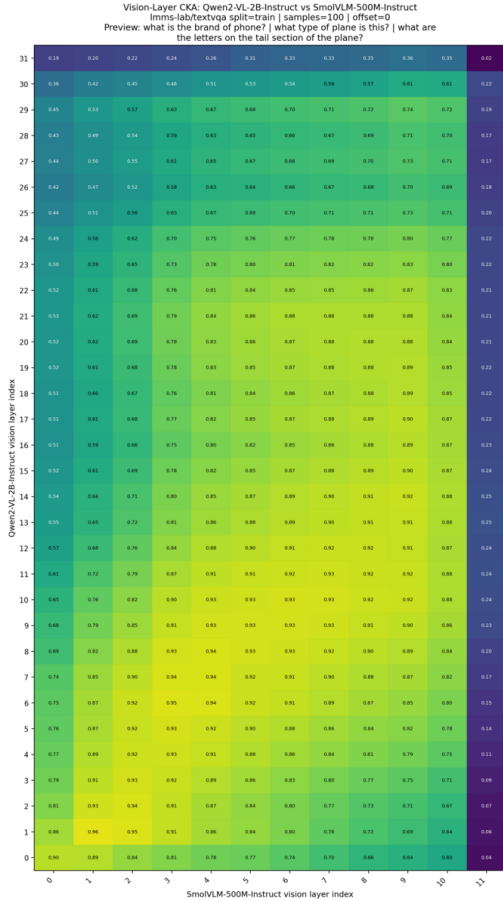


(a) TextVQA

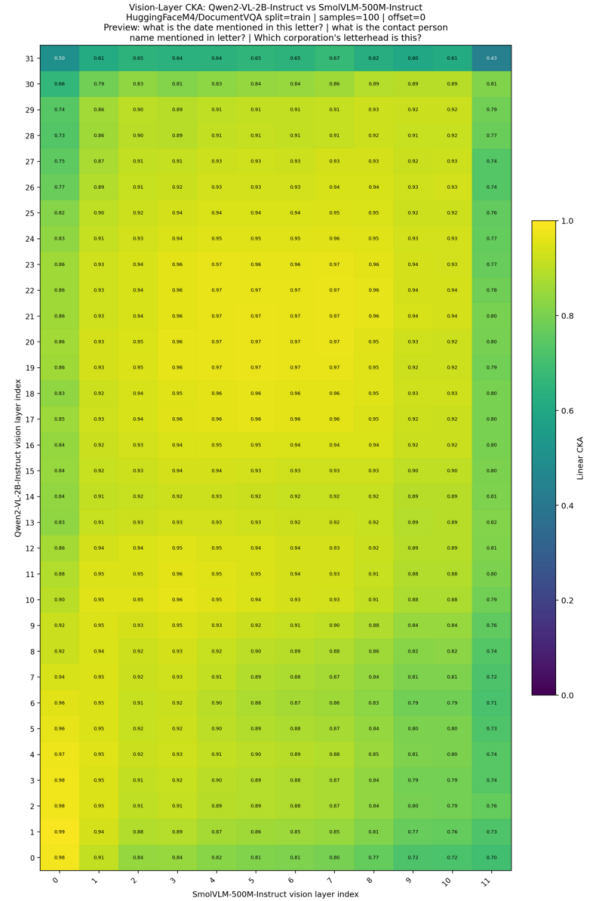


(b) DocVQA

Figure 5.6: Final-layer CKA heatmaps for Qwen2-VL-2B and SmolVLM-500M on TextVQA and DocVQA.



(a) TextVQA



(b) DocVQA

Figure 5.7: Vision-layer CKA heatmaps for Qwen2-VL-2B and SmolVLM-500M on TextVQA and DocVQA.

However, Tables 5.3, 5.4 and 5.5 still show a clear benchmark gap between the student models and stronger teachers. The layer-wise baseline also fails to consistently outperform output-level distillation.

This pattern supports the design choice in GRACE: output-level tokenizer-agnostic KD and sample-dependent teacher weighting are more reliable than representation matching alone, because high hidden-state similarity is not a reliable signal that direct layer matching will produce better task transfer.

5.3 Ablation Study

This section answers RQ2 from Chapter 1 by isolating the main components of GRACE on DocVQA. Table 5.8 keeps the teacher set and KD loss close to the final method and changes one mechanism at a time. The Cross-family ULD row uses one cross-family teacher and optimizes $\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD}\mathcal{L}_{ULD}$, with $\mathcal{L}_{ULD} = \|\pi(p_s) - \pi(p_t)\|_1$, providing the single-teacher tokenizer-mismatched baseline. Uniform averaging adds the full teacher pool but uses fixed equal weights:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD} \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{TW}^{(i)}.$$

The routing-only variant keeps sparse teacher selection but replaces post-routing weights with uniform weights, so

$$w_i^{(b)} = u_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b]}{|\mathcal{R}_b|}.$$

Full GRACE combines the three mechanisms and uses

$$w_i^{(b)} \propto (\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}}.$$

The $\lambda_{\text{blend}} = 1.0$ row keeps the GRACE active-set threshold and fallback rules, but removes the score-softmax factor from the confident-score case. In that case, the final weights reduce to router weights restricted to the active routed set:

$$w_i^{(b)} \propto \hat{p}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b].$$

Thus, this row is an active-set-restricted router-weight ablation, not a pure router-only ablation over all routed teachers. The $\beta_{\text{EMA}} = 0$ row removes temporal smoothing by setting $\hat{s}_i = s_i$ before score-softmax weighting.

Table 5.8 shows that the full method is the strongest setting for both student sizes. The ablation study highlights three findings about teacher diversity, routing and gradient-agreement refinement.

Table 5.8: DocVQA component ablation for SmolVLM-256M and SmolVLM-500M.

DocVQA ablation variant	SmolVLM-256M	SmolVLM-500M
Cross-family ULD (single teacher)	52.24	65.28
Uniform averaging	55.70	69.59
Routing only + trie-Wasserstein	57.45	63.23
Full GRACE + trie-Wasserstein	60.48	73.11
Full GRACE + trie-Wasserstein ($\lambda_{\text{blend}} = 1.0$)	57.97	70.45
Full GRACE + trie-Wasserstein ($\beta_{\text{EMA}} = 0$)	57.45	71.25

Teacher diversity is necessary but not sufficient because uniform multi-teacher averaging adds useful supervision while still failing to resolve tokenizer mismatch, teacher specialization and conflicting KD directions. Moving from a single cross-family teacher (ULD) to uniform four-teacher averaging adds +3.46 on SmolVLM-256M and +4.31 on SmolVLM-500M. The gain confirms that multiple heterogeneous teachers provide better supervision than any single teacher, but uniform weighting leaves a further gap of +4.78/+3.52 to the full method.

Routing without gradient refinement can hurt. On SmolVLM-500M, routing-only scoring (63.23) is *worse* than uniform averaging (69.59) by -6.36 points. The router selects teachers but assigns them incorrect relative weights. Without GRACE, these wrong weights degrade the result below a naive equal average. On SmolVLM-256M the pattern reverses: routing-only (57.45) beats uniform (55.70) by $+1.75$. The larger student extracts stronger routing features from its hidden states, so its router produces a sharper but less accurate ranking that amplifies errors when uncorrected. The SmolVLM-256M routing signal is closer to uniform, so the damage is smaller.

Gradient-agreement refinement acts as the critical corrective layer because it adjusts routed teacher weights according to whether each teacher’s KD direction agrees with the supervised task objective. Full GRACE recovers +9.88 points over routing-only on SmolVLM-500M and +3.03 on SmolVLM-256M. Decomposing the 500M gain, the active-set router-weight variant ($\lambda_{\text{blend}} = 1.0$) already recovers +7.22 points ($63.23 \rightarrow 70.45$) by replacing uniform post-routing weights with learned router weights while retaining the GRACE active-set filter. Full score-softmax blending then adds another +2.66 points ($70.45 \rightarrow 73.11$). On SmolVLM-256M, the active-set router-weight variant adds only +0.52 points ($57.45 \rightarrow 57.97$), while full score-based blending adds +2.51 points ($57.97 \rightarrow 60.48$). These results show that active-set filtering and router weighting help, but agreement-score weighting is still needed for the strongest final performance.

Removing EMA smoothing ($\beta_{\text{EMA}} = 0$) degrades both students (256M: $60.48 \rightarrow 57.45$, 500M: $73.11 \rightarrow 71.25$), confirming that temporal smoothing stabilizes score estimates but is secondary to the refinement step itself.

Answer to RQ2

The ablation isolates three cumulative contributions: (1) heterogeneous teacher diversity (+3 to +4 pts over single-teacher ULD), (2) sparse routing structure (beneficial only when paired with refinement) and (3) gradient-agreement refinement, which adds +3 to +10 pts over routing-only and is the single largest contributor. EMA smoothing provides additional stabilization by reducing the effect of noisy batch-level gradient-agreement scores before they are converted into final KD weights. Without gradient refinement, routing can degrade performance below uniform averaging.

5.4 Efficiency Analysis

This section addresses RQ3 from Chapter 1. Runtime, training cost and storage overhead are analyzed using the H200 rental rate and the observed Cloudflare R2 storage footprint.

Answer to RQ3

The cost of GRACE is dominated by GPU training, not object storage. From the Vast.ai screenshot, one H200 instance costs 2.059 USD/hour, so a 7-hour run costs about $7 \times 2.059 = 14.41$ USD and a 2500 USD budget corresponds to

$$\frac{2500}{2.059} \approx 1214.18 \text{ GPU-hours,}$$

which is about 50.59 days of continuous single-GPU runtime. If one full run takes about 7 hours, this budget is equivalent to roughly

$$\frac{1214.18}{7} \approx 173$$

single-H200 runs.

Storage costs are much lower under the official Cloudflare R2 standard pricing policy: 0.015 USD/GB-month for storage, 4.50 USD per million Class A operations, 0.36 USD per million Class B operations and zero egress. Using the dashboard values in Figure 4.2, 1.04 TB of stored logits is about 1040 GB, which gives

$$1040 \times 0.015 = 15.60 \text{ USD/month.}$$

The 598.73k Class A operations stay below the 1M monthly free allowance, so their marginal cost is 0 USD. The 20.3M Class B operations exceed the 10M free allowance by 10.3M, which gives

$$10.3 \times 0.36 = 3.71 \text{ USD}.$$

The total R2 cost is therefore about

$$15.60 + 3.71 = 19.31 \text{ USD/month},$$

before billing-unit rounding. This is below 1% of the 2500 USD GPU budget, which means the main efficiency bottleneck is training compute and offline logit generation, not storage egress or serving cost.

5.5 Discussion

The results show a clear ordering across method families. Same-family KL baselines are weakest, single cross-family distillation is stronger, uniform multi-teacher weighting improves further and GRACE is best overall [21, 25]. No single component fully addresses the heterogeneous distillation problem: heterogeneous teachers provide better supervision, but the student still needs a tokenizer-agnostic loss, sample-dependent routing and a stable post-routing weighting scheme.

The training recipe explains this pattern because GRACE combines a shared-trie Wasserstein loss for mismatched tokenizers with routed multi-teacher selection, top- k masking, routed-weight renormalization and EMA-smoothed gradient-agreement refinement [25, 38, 42]. The smoother DocVQA training curve in Figure 5.4 is consistent with that design: the router narrows the teacher set first, then GRACE reweights only the routed teachers, reducing the instability caused by uniformly mixing heterogeneous teacher signals. The gradient-agreement component can also be understood from a first-order multi-objective view. Let $g_{CE} = \nabla_{\theta} \mathcal{L}_{CE}$ and $g_i = \nabla_{\theta} \mathcal{L}_{KD}^{(i)}$ be the supervised and teacher-specific KD gradients. A combined update with teacher weights w_i is

$$\Delta\theta = -\eta \left(g_{CE} + \alpha_{KD} \sum_i w_i g_i \right).$$

The first-order change in the supervised loss is

$$\Delta \mathcal{L}_{CE} \approx g_{CE}^{\top} \Delta\theta = -\eta \|g_{CE}\|_2^2 - \eta \alpha_{KD} \sum_i w_i \langle g_{CE}, g_i \rangle.$$

If $\langle g_{CE}, g_i \rangle < 0$, then teacher i contributes a term that increases the first-order CE loss instead of reducing it. This is the gradient-conflict case studied in multi-task optimization [70]. GRACE does not project gradients as PCGrad does. Instead, it uses the cosine agreement

$$\cos(g_{CE}, g_i) = \frac{\langle g_{CE}, g_i \rangle}{\|g_{CE}\|_2 \|g_i\|_2}$$

as a teacher-weighting signal. This explains why post-routing gradient refinement can improve over routing alone.

The largest gain appears on ChartQA and the trend holds for SmolVLM-256M as well. Sample-dependent teacher assignment is most useful when the task mixes visual parsing and numerical reasoning and the method benefits both evaluated student models.

The comparison with SHM-CKA reinforces this conclusion by showing that heterogeneous teacher access is not enough when the method relies only on hidden-state matching instead of routed output-level KD. SHM-CKA is competitive on ChartQA yet clearly weaker on TextVQA and DocVQA. Together with the CKA heatmaps, this confirms that internal similarity alone does not guarantee effective transfer. **Sample-dependent teacher weighting therefore remains necessary, particularly when the teacher pool contains models with different tokenizers, architectures and task-specific strengths** [27, 39].

5.6 Limitations and Threats to Validity

Cost-limited evaluation. Because the reported numbers are single-run results, the comparison does not provide confidence intervals or repeated-seed estimates for small gaps between strong baselines. A typical H200 run takes about 7 hours and costs about 14.41 USD at 2.059 USD/hour on Vast.ai before offline cache-generation and storage cost, which limits repeated seeds and broader hyperparameter sweeps.

Limited sweep coverage. The main tables use a shared four-teacher pool for Uniform, SHM-CKA, REINFORCE and GRACE, but the study still reports single completed runs and does not exhaustively sweep seeds, teacher subsets or hyperparameters. Although this does not invalidate the overall trend, it limits the strength of strict uncertainty claims about closely matched methods.

Answer-token-centric transfer. The method distills teacher distributions on supervised answer positions, routes teachers from pooled student states and refines them with CE/KD gradient agreement. Because it does not explicitly align visual tokens, cross-attention maps or region-level evidence, some gains may come mainly from output-side transfer instead of deeper visual grounding.

Systems dependence. The implementation relies on large offline teacher-logit caches, remote object storage and local streaming caches. Although this design keeps the online loop simple, it increases storage, I/O and reproducibility requirements because training depends on large offline teacher-logit caches and remote streaming infrastructure. The method is therefore easier to reproduce in a well-provisioned training environment than on minimal local hardware.

Scope of validation. The empirical study focuses on VQA benchmarks and SmolVLM-family students. Although the reported gains are strong in this regime, the thesis does not yet establish whether the same behavior holds for other multimodal task families, other student architectures or larger teacher pools.

Conclusion and Future Work

This thesis studied vision-language model distillation under heterogeneous teachers, where teacher utility varies across samples and teacher-student tokenizers do not match. To address these challenges, we presented Gradient-Routed Alignment with Cross-tokenizer Experts (GRACE), a routed multi-teacher framework with trie-Wasserstein tokenizer-agnostic distillation, sparse top- k selection and gradient-agreement refinement. The training procedure routes teachers first, refines their weights with gradient-agreement scores and then aggregates the routed trie-Wasserstein KD loss [25, 38, 42].

Empirically, GRACE obtains the highest scores among the methods compared in the main tables, indicating that selective routing and post-routing gradient refinement improve heterogeneous teacher-to-student VLM distillation under the evaluated VQA setting. For SmolVLM-500M, it reaches 69.38 on TextVQA, 73.11 on DocVQA and 80.20 on ChartQA. For SmolVLM-256M, GRACE reaches 60.73 on TextVQA, 60.48 on DocVQA and 72.32 on ChartQA, showing that the method improves both evaluated student models. These results exceed the original students, direct finetuning and all reported distillation baselines in Chapter 5. On SmolVLM-500M, GRACE closes about 55% of the original teacher-student gap on TextVQA and ChartQA and about 40% on DocVQA. The results confirm that combining selective routing with post-routing score refinement improves heterogeneous teacher-to-student VLM distillation under the benchmarks and teacher pool studied here.

Future Work

Future work should investigate three directions that would strengthen the empirical scope, reduce systems cost and extend the method beyond answer-token distillation. First, **stronger evaluation**: repeated runs with multiple seeds would tighten the comparison, especially for the smaller gaps between close multi-teacher baselines. Second, **systems efficiency**: the pipeline still relies on large offline teacher-logit caches, so reducing storage and I/O cost would make the method easier to scale to more datasets and larger teacher sets. Third, **broader method extensions**: the current method remains answer-token-centric, so one important next step is explicit visual distillation through visual-token alignment, cross-attention transfer, or region-level matching. Future work can also explore richer router inputs, improved objective-conflict handling and teacher sets that mix specialized document, chart and general VLM experts [33, 34, 40].

Chapter A

Closed-Form Trie-Wasserstein Derivation

This appendix derives the closed-form expression used by the KD loss in Chapter 3. The result is stated for a general rooted tree [26, 38].

A.1 Setup

Let $T = (V, E, r)$ be a rooted tree with root r . Each edge $e \in E$ has a positive weight $w(e) > 0$. For an edge $e = (p(e), c(e))$, where $p(e)$ is the parent and $c(e)$ is the child, let

$$\Gamma(e) = \{v \in V \mid e \text{ lies on the path from } r \text{ to } v\}$$

denote the subtree below edge e .

For two vertices $u, v \in V$, define the tree metric

$$d_T(u, v) = \sum_{e \in \text{path}(u, v)} w(e),$$

where $\text{path}(u, v)$ is the unique set of edges on the path between u and v .

Let μ and ν be two probability measures on V . In the trie-Wasserstein setting, these correspond to student and teacher token distributions placed on the leaves of the shared byte trie. Measures on leaves can be viewed as measures on all vertices by assigning zero mass to non-leaf vertices.

The Kantorovich form of the W_1 optimal transport distance under the tree metric is [38]

$$W_{1,T}(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \sum_{u \in V} \sum_{v \in V} \pi(u, v) d_T(u, v),$$

where $\Pi(\mu, \nu)$ is the set of couplings satisfying

$$\sum_{v \in V} \pi(u, v) = \mu(u), \quad \sum_{u \in V} \pi(u, v) = \nu(v), \quad \pi(u, v) \geq 0.$$

For each edge e , define the subtree-mass imbalance

$$\Delta(e) = \mu(\Gamma(e)) - \nu(\Gamma(e)).$$

We prove that

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\Delta(e)|.$$

A.2 Lower Bound

Consider any feasible transport plan $\pi \in \Pi(\mu, \nu)$. Removing an edge e separates the tree into two components: the subtree $\Gamma(e)$ and its complement $V \setminus \Gamma(e)$.

The total source mass initially inside $\Gamma(e)$ is $\mu(\Gamma(e))$, while the total target mass required inside $\Gamma(e)$ is $\nu(\Gamma(e))$. If

$$\mu(\Gamma(e)) > \nu(\Gamma(e)),$$

then at least $\mu(\Gamma(e)) - \nu(\Gamma(e))$ mass must leave $\Gamma(e)$. If

$$\mu(\Gamma(e)) < \nu(\Gamma(e)),$$

then at least $\nu(\Gamma(e)) - \mu(\Gamma(e))$ mass must enter $\Gamma(e)$. Therefore, any feasible plan must move at least

$$|\Delta(e)| = |\mu(\Gamma(e)) - \nu(\Gamma(e))|$$

mass across edge e .

Let $q_\pi(e)$ be the amount of mass transported by π whose path crosses edge e . The argument above implies

$$q_\pi(e) \geq |\Delta(e)|.$$

Because the cost of moving mass from u to v is the sum of edge weights along the unique path between them, the total cost of π can be decomposed edgewise:

$$\sum_{u,v \in V} \pi(u,v) d_T(u,v) = \sum_{e \in E} w(e) q_\pi(e).$$

Using $q_\pi(e) \geq |\Delta(e)|$, we obtain

$$\sum_{u,v \in V} \pi(u,v) d_T(u,v) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

Since this holds for every feasible transport plan π , it also holds for the optimal plan:

$$W_{1,T}(\mu, \nu) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

A.3 Achievability

It remains to show that the lower bound can be achieved.

Define a signed measure

$$\sigma(v) = \mu(v) - \nu(v).$$

For each edge e , the subtree imbalance is

$$\Delta(e) = \sum_{v \in \Gamma(e)} \sigma(v).$$

Interpret $\Delta(e)$ as the net amount of mass that must cross edge e . If $\Delta(e) > 0$, the subtree below e contains surplus mass and $\Delta(e)$ mass must move from $\Gamma(e)$ to its complement. If $\Delta(e) < 0$, the subtree has a deficit and $|\Delta(e)|$ mass must enter $\Gamma(e)$.

Because T is a tree, there is a unique path between any two vertices. A bottom-up transport construction can move surplus mass upward from subtrees with $\Delta(e) > 0$ and move mass downward into subtrees with $\Delta(e) < 0$. This construction sends exactly $|\Delta(e)|$ mass across each edge e . It satisfies all supply and demand constraints because the total signed mass is zero:

$$\sum_{v \in V} \sigma(v) = \sum_{v \in V} \mu(v) - \sum_{v \in V} \nu(v) = 1 - 1 = 0.$$

The cost of this construction is therefore

$$\sum_{e \in E} w(e) |\Delta(e)|.$$

Thus there exists a feasible transport plan with cost equal to the lower bound. Hence

$$W_{1,T}(\mu, \nu) \leq \sum_{e \in E} w(e) |\Delta(e)|.$$

Combining this upper bound with the lower bound gives

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|.$$

References

- [1] Wikimedia Commons contributors. File: Contrastive language-image pretraining.png. Wikimedia Commons, 2024. MIT License. Accessed: 2026-03-06.
- [2] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021.
- [3] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning, 2022.
- [4] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, 2023.
- [5] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning, 2023.
- [6] Yushi Hu, Otilia Stretcu, Chun-Ta Lu, Krishnamurthy Viswanathan, Kenji Hata, Enming Luo, Ranjay Krishna, and Ariel Fuxman. Visual program distillation: Distilling tools and programmatic reasoning into vision-language models, 2024.
- [7] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read, 2019.
- [8] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. Docvqa: A dataset for vqa on document images, 2021.
- [9] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning, 2022.
- [10] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces, 2024.

- [11] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020.
- [12] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022.
- [13] Xiangxiang Chu, Limeng Qiao, Xinyang Lin, Shuang Xu, Yang Yang, Yiming Hu, Fei Wei, Xinyu Zhang, Bo Zhang, Xiaolin Wei, and Chunhua Shen. Mobilevlm: A fast, strong and open vision language assistant for mobile devices, 2023.
- [14] Baichuan Zhou, Ying Hu, Xi Weng, Junlong Jia, Jie Luo, Xien Liu, Ji Wu, and Lei Huang. Tinyllava: A framework of small-scale large multimodal models, 2024.
- [15] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, page 5998–6008, 2017.
- [16] Andrés Marafioti, Orr Zohar, Miquel Farré, Merve Noyan, Elie Bakouch, Pedro Cuenca, Cyril Zakka, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, Vaibhav Srivastav, Joshua Lochner, Hugo Larcher, Mathieu Morlon, Lewis Tunstall, Leandro von Werra, and Thomas Wolf. Smolvlm: Redefining small and efficient multimodal models, 2025.
- [17] Andrés Marafioti, Merve Noyan, Miquel Farré, Elie Bakouch, and Pedro Cuenca. Smolvlm - small yet mighty vision language model. Hugging Face Blog, 2024. Accessed: 2026-04-02.
- [18] AI Rank. Textvqa benchmark: Complete leaderboard and performance analysis. AI Rank, 2026. Self-reported leaderboard. Accessed: 2026-04-02.
- [19] AI Rank. Docvqa benchmark: Complete leaderboard and performance analysis. AI Rank, 2026. Self-reported leaderboard. Accessed: 2026-04-02.
- [20] AI Rank. Chartqa benchmark: Complete leaderboard and performance analysis. AI Rank, 2026. Self-reported leaderboard. Accessed: 2026-04-02.

- [21] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015.
- [22] Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. A survey on knowledge distillation of large language models, 2024.
- [23] Zi Wang. Zero-shot knowledge distillation from a decision-based black-box model. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, page 10675–10685. PMLR, 2021.
- [24] Wenhao Ma, Jie Li, Bo Han, and Xiaojuan Qi. B2kd: A unified framework for knowledge distillation from black-box models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 15313–15323, 2024.
- [25] Nicolas Boizard, Kevin El Haddad, Pierre Colombo, and Chloé Clavel. Towards cross-tokenizer distillation: The universal logit distillation loss for llms, 2024.
- [26] Yuki Takezawa, Ryoma Sato, and Makoto Yamada. Supervised tree-wasserstein distance. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, page 10086–10095. PMLR, 2021.
- [27] Sayantan Dasgupta and Trevor Cohn. Improving language model distillation through hidden state matching. In *International Conference on Learning Representations*, 2025.
- [28] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. Gptq: Accurate post-training quantization for generative pre-trained transformers, 2022.
- [29] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. Smoothquant: Accurate and efficient post-training quantization for large language models. In *Proceedings of the 40th International Conference on Machine Learning*, page 38087–38099, 2023.
- [30] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. Awq: Activation-aware weight quantization for llm compression and acceleration, 2024.

- [31] Shiyao Li, Yingchun Hu, Xuefei Ning, Xihui Liu, Ke Hong, Xiaotao Jia, Xiuhong Li, Yaqi Yan, Pei Ran, Guohao Dai, Shengen Yan, Huazhong Yang, and Yu Wang. Mbq: Modality-balanced quantization for large vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, page 4167–4177, 2025.
- [32] Wikimedia Commons contributors. File: Knowledge-distillation-example.png. Wikimedia Commons, 2023. CC0 1.0. Source: Intel Distiller knowledge distillation page. Accessed: 2026-03-07.
- [33] Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, Ming Gong, Yan Fu, and Daxin Jiang. Reinforced multi-teacher selection for knowledge distillation, 2020.
- [34] Hailin Zhang, Defang Chen, and Can Wang. Adaptive multi-teacher knowledge distillation with meta-learning, 2023.
- [35] Zhiyuan Fang, Jianfeng Wang, Xiaowei Hu, Lijuan Wang, Yezhou Yang, and Zicheng Liu. Compressing visual-linguistic model via knowledge distillation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, page 1428–1438, 2021.
- [36] Kaicheng Yang, Tiancheng Gu, Xiang An, Haiqiang Jiang, Xiangzi Dai, Ziyong Feng, Weidong Cai, and Jiankang Deng. Clip-cid: Efficient clip distillation via cluster-instance discrimination, 2024.
- [37] Tuan Le, Makoto Yamada, Kenji Fukumizu, and Marco Cuturi. Tree-sliced variants of wasserstein distances. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [38] Gabriel Peyré and Marco Cuturi. Computational optimal transport, 2019.
- [39] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, page 3519–3529. PMLR, 2019.
- [40] Yu-Chu Yu, Chi-Pin Huang, Jr-Jen Chen, Kai-Po Chang, Yung-Hsuan Lai, Fu-En Yang, and Yu-Chiang Frank Wang. Select and distill: Selective dual-teacher knowledge transfer for continual learning on vision-language models, 2024.

- [41] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, 2017.
- [42] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022.
- [43] Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Zhao, Andrew Dai, Zhifeng Chen, Quoc Le, and James Laudon. Mixture-of-experts with expert choice routing, 2022.
- [44] Yichen Zhu and Yi Wang. Student customized knowledge distillation: Bridging the gap between student and teacher. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, page 5057–5066, 2021.
- [45] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021.
- [46] Wikimedia Commons contributors. File: Transformer, attention block diagram.png. Wikimedia Commons, 2024. CC BY 4.0. Accessed: 2026-03-06.
- [47] Wikimedia Commons contributors. File: Vision transformer.png. Wikimedia Commons, 2024. CC BY 4.0. Accessed: 2026-03-06.
- [48] Jianhua Lin. Divergence measures based on the shannon entropy. *IEEE Transactions on Information Theory*, 37(1), 1991.
- [49] Rémi Flamary and Nicolas Courty. Optimal transport for machine learning: Introduction to optimal transport, 2019. Tutorial at ISBI 2019.
- [50] Wikimedia Commons contributors. File: Optimal transport matrix.png. Wikimedia Commons, 2018. CC BY-SA 4.0. Accessed: 2026-03-31.
- [51] DeepSeek-AI. Deepseek-v2: A strong, economical and efficient mixture-of-experts language model, 2024.
- [52] Wikimedia Commons contributors. File: Deepseek moe and mla (deepseek-v2).svg. Wikimedia Commons, 2025. MIT License. Accessed: 2026-03-06.
- [53] Haodong Duan, Xinyu Fang, Junming Yang, et al. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models, 2024.

- [54] Wikimedia Commons contributors. File: Stop sign.jpg. Wikimedia Commons, 2006. Public domain. Accessed: 2026-03-06.
- [55] Wikimedia Commons contributors. File: Receiptswiss.jpg. Wikimedia Commons, 2015. CC BY-SA 3.0. Accessed: 2026-03-06.
- [56] Wikimedia Commons contributors. File: Bar chart.svg. Wikimedia Commons, 2016. CC BY-SA 4.0. Accessed: 2026-03-06.
- [57] Samyam Rajbhandari, Jeff Rasley, Olatunji Ruwase, and Yuxiong He. Zero: Memory optimizations toward training trillion parameter models, 2020.
- [58] Tianqi Chen, Bing Xu, Chiyuan Zhang, and Carlos Guestrin. Training deep nets with sublinear memory cost, 2016.
- [59] Hugging Face TB. Smolvlm2-2.2b-instruct model card. Hugging Face, 2025. Accessed: 2026-04-02.
- [60] Yifan Cui, Wei Wang, Yifei Wang, Yao Lu, Haoyu Wang, Fuzhen Zhuang, Qing He, and Chuan Shi. Sinkhorn distance minimization for knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, page 11046–11060. Association for Computational Linguistics, 2024.
- [61] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3):229–256, 1992.
- [62] Yuqiao Wen, Zichao Li, Wenyu Du, and Lili Mou. f-divergence minimization for sequence-level knowledge distillation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 10817–10834. Association for Computational Linguistics, 2023.
- [63] Cloudflare. R2 object storage. Cloudflare developer documentation, 2026. Accessed: 2026-04-09.
- [64] Cloudflare. Use the aws sdk for python (boto3) with r2. Cloudflare developer documentation, 2026. Accessed: 2026-04-09.
- [65] fsspec developers. Features. fsspec documentation, 2026. Accessed: 2026-04-09.
- [66] fsspec developers. fsspec.implementations.cached. fsspec documentation, 2026. Accessed: 2026-04-09.

- [67] Walid Bousselham, Rishikesh Jadhav, Simon de Mathelin, François Fleuret, and Pascal Frossard. Dex-ar: Explainability in multimodal large language models with deep expectation and attention rollout, 2026.
- [68] Taiqiang Wu, Chaofan Tao, Jiahao Wang, Runming Yang, Zhe Zhao, and Ngai Wong. Rethinking kullback-leibler divergence in knowledge distillation for large language models. In *Proceedings of the 31st International Conference on Computational Linguistics*, page 5737–5755, 2025.
- [69] Max Klabunde, Tobias Schumacher, Markus Strohmaier, and Florian Lemmerich. Similarity of neural network models: A survey of functional and representational measures, 2023.
- [70] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, page 5824–5836, 2020.